CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

In many educational institutions, departments are allotted budgets every academic year for organizing academic activities such as workshops, seminars, conferences, and other departmental events. Managing these budgets manually often becomes difficult because expenses are recorded in different files or registers, which may lead to errors and lack of proper tracking. Without a centralized system, it is challenging for departments to monitor how much of the allocated budget has been used and how much remains available. This situation can sometimes result in overspending or delays in approving important expenses.

The Department Budget Tracking System is developed using the Salesforce platform to provide a centralized and organized way of managing departmental budgets and expenses. The proposed system allows departments to allocate budgets, record expenses, and monitor budget utilization in real time. Custom objects such as Department, Budget Allocation and Expense are used to store and manage the required data. When an expense is entered into the system, the corresponding budget is automatically updated by deducting the amount from the recurring or non recurring budget category.

The proposed system also includes an approval mechanism for higher expenses and provides role based access for different users such as Faculty members, Head of Department (HOD), and Budget Administrators. Through Salesforce Experience Cloud, users can access the system through a web portal to submit expenses, review approvals and monitor budget usage. In addition, the system generates reports and dashboards that help administrators understand monthly expenses, track department spending and identify when the budget usage exceeds certain limits. This approach improves transparency, accountability and efficient financial management within the institution.

1.2 EXISTING SYSTEM

In many educational institutions, department budgets and expenses are often managed using manual methods such as spreadsheet tools. Departments receive budgets for academic activities like workshops, seminars, conferences and other events. These expenses are commonly recorded using spreadsheet applications such as Microsoft Excel or Google Sheets. While these tools help store financial information in a structured format they mainly depend on manual data entry and updates.

The existing system used in the department in budget allocation and expense tracking are handled through spreadsheets maintained by administrators or faculty members. Budget details, expense records and remaining balances are manually entered and calculated in the spreadsheet. Faculty members submit expense details to the department, and the administrator updates the spreadsheet accordingly. Although this method helps maintain basic financial records, it lacks automation and real time monitoring.

Since spreadsheets require manual calculations and updates there is a high possibility of errors in budget calculations or data entry. It also becomes difficult to track the exact budget utilization and remaining balance in real time. When multiple expenses are recorded, maintaining accurate financial data and verifying expense approvals becomes more complicated. In addition, spreadsheets do not provide automated notifications or structured approval workflows for managing expenses.

Another limitation of the existing system is the lack of centralized access and data security. When multiple users update the spreadsheet, there may be issues such as duplicate entries, data inconsistencies and version control problems. Reports and analysis must also be generated manually, which requires additional effort and time. Because of these limitations, the existing spreadsheet based system is not efficient for managing departmental budgets and monitoring expenses effectively.

Therefore there is a need for a centralized and automated system that can manage budget allocation, expense tracking and approval workflows more efficiently. A dedicated system can help departments monitor budget usage, maintain accurate records and provide better financial insights for decision making.This system can also improve transparency by allowing authorized users to access and monitor budget information through a centralized platform. It can reduce manual workload by automating calculations, notifications and approval processes. Additionally, the system can generate reports and dashboards that help administrators easily analyze department expenses and budget utilization.

1.3 DRAWBACKS OF EXISTING SYSTEM

1. Many financial management applications are designed for large organizations and are not specifically tailored for departmental budget tracking in educational institutions.
2. In many institutions, budget data is still maintained using spreadsheets, which require manual updates and calculations.
3. Spreadsheet based systems depend on manual calculations and updates which may lead to errors and make it difficult to accurately track department expenses and remaining budget.
4. Approval of high value expenses is often handled through paperwork, which may delay the approval process.
5. Difficulty in tracking how much of the allocated budget has already been used and how much remains available.

1.4 PROPOSED SYSTEM

The proposed system, Department Budget Tracking System using Salesforce Experience Cloud is developed to manage and monitor the annual budget of the MCA Department in an organized and transparent manner. The system provides a centralized platform where the department can maintain budget details, record expenses and track the utilization of funds for different academic activities such as seminars, workshops, symposiums and other departmental programs.

In the proposed system, the MCA department is assigned a budget for each academic year and all financial transactions related to departmental activities are recorded within the system. Custom objects such as Department, Budget Allocation and Expense are used to manage the data. When an expense is entered, the system automatically deducts the amount from the allocated budget and updates the remaining balance. The budget can be categorized as recurring or non recurring, which helps the department track different types of expenses separately.

The proposed system also includes an Role-based access is provided for different users such as Faculty members, HOD and Budget Administrator, allowing them to perform their respective tasks within the system. Faculty members can submit expenses, the budget admin, the HOD can monitor the overall budget details also can review and approve them.

Through the Salesforce Experience Cloud portal, authorized users can access the system to submit expenses, check the remaining budget and view approval status. The system also generates monthly and year wise reports and dashboards that help the department analyze total expenses, remaining budget and spending patterns for each academic year. Email notifications are sent when an expense requires approval or when the budget usage crosses a certain limit. This system improves transparency, reduces manual work and helps the MCA department manage its yearly budget more efficiently.

1.5 ADVANTAGES OF PROPOSED SYSTEM

1. A centralized digital platform is developed to manage the MCA department budget and expenses efficiently.
2. Faculty members can easily submit expense requests through a simple and user friendly system.
3. The remaining budget is automatically updated whenever a new expense is recorded, reducing manual calculations and errors.
4. An approval workflow is implemented where the HOD can review and approve expenses digitally.
5. Reports and dashboards help administrators analyze budget usage and monitor departmental financial activities in real time.

SUMMARY

This chapter introduced the concept of the Department Budget Tracking System using Salesforce Experience Cloud developed for managing the MCA department's yearly budget. It explained the limitations of existing systems and spreadsheet tools, which are often complex or not suitable for tracking budgets and expenses at the department level.The drawbacks of these systems highlighted the need for a centralized platform to track departmental expenses efficiently. The proposed system provides an organized digital solution where faculty members can submit expenses, the Budget Admin can review, the HOD can monitor overall budget utilization and approve requests. The detailed analysis of the project will be discussed in Chapter 2.

CHAPTER 2

SYSTEM ANALYSIS

2.1 IDENTIFICATION OF NEED

Many educational institutions manage departmental budgets and expenses using manual methods such as spreadsheet tools like Microsoft Excel or Google Sheets. In these systems, budget allocations and expense details are recorded manually in spreadsheets maintained by administrators or faculty members. Although spreadsheets help store financial information, they depend heavily on manual data entry and calculations, which may lead to errors and data inconsistencies. It also becomes difficult to track the exact budget utilization and remaining balance in real time.

Departments such as the MCA department require a simpler and more organized system to manage department level budgets, expense submissions and approval processes. Faculty members need an easy way to submit expenses related to seminars, workshops and departmental events, while administrators require a clear view of how much budget has been used and how much remains available. Manual spreadsheet systems do not provide automated approval workflows or real time monitoring of expenses.

Therefore, there is a need for a centralized and user friendly system that can efficiently manage departmental budgets and track expenses. The proposed Department Budget Tracking System using Salesforce Experience Cloud provides a structured platform for budget allocation, expense submission, approval management and financial monitoring within the MCA department. This system helps improve accuracy, transparency and efficient management of departmental financial activities.

2.2 FEASIBILITY STUDY

A feasibility study is an evaluation of the practicality and viability of a proposed system before its development and implementation. It helps determine whether the project can be successfully developed within the available resources and whether it will meet the required objectives. The proposed Department Budget Tracking System using Salesforce Experience Cloud has the feasibility study, that helps determine whether the system can efficiently manage departmental budgets, track expenses and provide better financial monitoring within the MCA department. The system analysis includes collecting and evaluating information related to the system and its environment. The feasibility study is generally categorized into the following types:

1. Technical feasibility
2. Economic feasibility
3. Operational feasibility

2.2.1 Technical Feasibility

Technical feasibility focuses on evaluating whether the proposed system can be developed using the available technology and technical resources. The proposed Department Budget Tracking System is developed using the Salesforce platform and Experience Cloud, which provide a secure and scalable environment for building enterprise applications. Salesforce allows developers to create custom objects, workflows, approval processes, and dashboards that help manage departmental budgets and expenses efficiently.

Technical feasibility focuses on evaluating whether the proposed system can be developed using the available technology and technical resources. Salesforce allows developers to create custom objects, workflows, approval processes, and dashboards that help manage departmental budgets and expenses efficiently. Since Salesforce is a reliable cloud platform with built-in security, scalability, and data management capabilities, the development and implementation of the proposed system are technically feasible.

2.2.2 Economic Feasibility

Economic feasibility evaluates whether the proposed system is financially beneficial and whether the benefits of the system outweigh the development and operational costs. The Department Budget Tracking System is developed using the Salesforce platform, which reduces the need for additional infrastructure or hardware. Since the system is cloud-based, maintenance and operational costs are minimal.

The system helps reduce manual work involved in maintaining budget records and calculating expenses. It also minimizes the chances of errors in financial records, which can improve the efficiency of budget management. By providing automated expense tracking, approval workflows, and real-time financial reports, the system helps administrators make better financial decisions.

Economic feasibility also considers the long-term benefits of the system, such as improved transparency, better financial monitoring, and efficient management of departmental budgets. These benefits contribute to better resource utilization and improved operational efficiency within the department. Therefore, the proposed system is economically feasible and beneficial for the MCA department.

2.2.3 Operational Feasibility

Operational feasibility determines whether the proposed system will be accepted and effectively used by the intended users. The Department Budget Tracking System is designed to be simple and user friendly so that faculty members, the Head of Department and the Budget Admin can easily interact with the system. Faculty members can submit expense requests related to departmental activities, while the Budget Admin can review and approve these requests.

The system also allows the Head of Department to view and monitor departmental expenses, helping ensure transparency in financial management. Through the Experience Cloud portal, users can easily access the system, submit information and monitor budget details without requiring extensive technical knowledge. The user interface is designed to be clear and organized so that users can quickly understand and use the system. Since the system simplifies budget management, reduces manual work and provides better financial monitoring, it is expected to be accepted and effectively used by the MCA department staff. Therefore the proposed system is operationally feasible.

2.3 SOFTWARE REQUIREMENT SPECIFICATIONS

Software Requirement Specification describes the functional and non functional requirements needed for the successful development and operation of the proposed system. It provides a clear understanding of the system features, system behavior and user interactions. The SRS helps developers and users understand how the system will function and ensures that the system meets the requirements of the organization.

The Department Budget Tracking System using Salesforce Experience Cloud is designed to manage departmental budgets and expenses efficiently. The system allows faculty members to submit expense requests, enables the Budget Admin to review and allows the HOD to monitor departmental spending and approve expenses. The system also maintains records of budget allocation and expense details and generates reports and dashboards to analyze departmental financial activities.

2.3.1 Hardware Requirements

The hardware requirements define the minimum physical components needed to operate the proposed system efficiently. Since the Department Budget Tracking System is developed using the Salesforce cloud platform, it does not require high end hardware. A basic computer system with internet connectivity is sufficient to access and use the application through a web browser. The following are the minimum hardware requirements required to run the system smoothly.

Processor	:	12th Gen Intel(R) Core(TM) i3-1215U 1.20 GHz
RAM	:	8 GB
Operating System	:	Windows 11
Hard Disk	:	160GB
System Type	:	64 bit Operating System, x64 based processor

2.3.2 Software Requirements

The software requirements describe the tools and platforms required for developing and operating the proposed system. The Department Budget Tracking System is developed using the Salesforce platform, which provides a cloud based environment for application development, data management and system access. The following software components are used for the development and operation of the system.

Platform	:	Salesforce
Development Tool	:	Visual Studio Code with Salesforce Extensions
Programming Language	:	Apex
User Interface	:	Lightning Experience
Portal Technology	:	Salesforce Experience Cloud
Database	:	Salesforce Data Storage (Custom Objects)

2.4 SOFTWARE DESCRIPTION

Software description explains the technologies and tools used for developing and operating the proposed system. It provides details about the front end, back end and database used in the application. The Department Budget Tracking System is developed using the Salesforce platform, which provides a secure cloud environment for building, managing and accessing applications. Salesforce supports custom object creation, workflow automation, approval processes and data management, making it suitable for managing departmental budgets and expenses efficiently. The system allows users such as faculty members, HOD and budget administrators to access the platform through a web based interface. It helps maintain departmental budget records, track expenses, and monitor financial activities through reports and dashboards.

2.4.1 Front End

The front end of the proposed system is developed using Salesforce Lightning Experience and Lightning Web Components (LWC). The front end represents the user interface of the application through which users interact with the system. Lightning provides a modern and responsive interface that improves the usability and accessibility of the application.

Lightning Web Components are reusable components that help create interactive pages for performing different operations in the system. Using these components, users can easily navigate through different sections such as department details, budget allocation, expense submission and report viewing. The front end is designed to be user friendly so that faculty members can submit expense requests, administrators can manage budget records and the Head of Department can monitor departmental expenses without requiring technical knowledge.

The interface also provides dashboards and reports that visually represent budget usage and expense details. These visual elements help users quickly understand financial data and make better decisions regarding departmental spending.

2.4.2 Back End

The back end of the proposed system is developed using Apex, the programming language provided by Salesforce. Apex is used to implement the core business logic of the system and manage the processing of data within the application. It helps handle operations such as expense submission, approval processing and automatic budget updates.

Salesforce automation tools such as workflow rules, approval processes, validation rules and triggers are used in the backend to automate system activities. For example when a faculty member submits an expense request, the system automatically sends a notification to the Budget Admin for approval. Once the expense is approved, the remaining budget amount is updated automatically.

The backend ensures that all system processes run efficiently and securely. It manages user permissions, handles data processing and maintains system integrity. These backend features help maintain accuracy and reliability in managing departmental financial records.

2.4.3 Database

The database used in the proposed system is Salesforce Data Storage, which stores data in the form of custom objects and records. Objects such as Department, Budget Allocation and Expense are created to store departmental budget information and expense details.

Each object contains fields that store important information such as department name, budget year, allocated amount, expense amount and approval status. Salesforce database provides secure data storage, easy data retrieval and efficient data management. It also supports reporting and dashboard features that allow administrators to analyze departmental budget usage and financial activities effectively.

SUMMARY

This chapter discussed the system analysis of the proposed Department Budget Tracking System using Salesforce Experience Cloud. It explained the need for the system and evaluated its feasibility in terms of economic, operational and technical aspects. The chapter also described the software requirement specifications, including the hardware & software required for the system and explained the technologies used such as the front end, back end and database. The next chapter discusses the system design of the proposed system.

CHAPTER 3

SYSTEM DESIGN

3.1 MODULE DESCRIPTION

A module description provides a detailed explanation of a particular component or feature within a system. It describes the purpose, functionality and operations performed by each part of the system. Modules help divide the entire system into smaller functional units so that each module handles a specific task efficiently. This approach improves system organization, simplifies development and makes the system easier to maintain. The following are the modules included in the Department Budget Tracking System.

1. Proposal Module
2. Budget Allocation Module
3. Expense Module
4. Reports and Dashboard Module

3.1.1 Proposal Module

The Proposal Module is used to manage the initial budget proposal submitted for the department. In the proposal module, the Budget Admin receives and reviews the proposed budget details required for departmental activities during the academic year. The proposal may include information such as the required budget amount and the type of budget needed for different academic events and activities.

The Proposal module helps in organizing and reviewing the proposed budget before it is officially allocated to the department. By maintaining the proposal details in a structured format, the system ensures that the budget planning process is clear and well organized. It also helps administrators verify the requested budget before moving to the allocation stage. The module also allows administrators to store proposal records for future reference and planning.

3.1.2 Budget Allocation Module

The Budget Allocation Module is used to assign the approved budget to the department for a specific academic year. Through this module, the Budget Admin can allocate a certain amount of budget and categorize it into recurring and non recurring budgets. This classification helps manage different types of departmental expenses efficiently.

The module maintains records of the total allocated budget and ensures that financial resources are properly distributed for departmental activities. It also allows administrators to monitor the available budget and maintain proper control over departmental spending. This module plays an important role in financial planning and budget management within the department.

3.1.3 Expense Module

The Expense Module allows faculty members to enter and submit expenses related to departmental activities such as workshops, seminars, conferences and other academic programs. Faculty members can provide details such as expense name, expense amount, date, category and purpose while submitting the expense information.

The expense module records all expenses and links them with the allocated departmental budget. Once an expense is entered, the system automatically updates the remaining budget amount. By digitizing the expense recording process, the module helps maintain accurate financial records and reduces the chances of manual errors.

3.1.4 Report and Dashboard Module

The Report and Dashboard Module generates various reports related to departmental budgets and expenses. These reports help administrators analyze financial activities and understand spending patterns within the department.

The module provides monthly and yearly reports showing total expenses, remaining budget and overall budget utilization. Dashboards visually represent financial data through charts and graphs, making it easier for administrators to understand the department's financial status. This module supports better decision making and helps maintain transparency in departmental financial management.

3.2 DATA FLOW DIAGRAM

A data flow diagram (DFD) is like a map that shows how information moves through a system. It uses symbols to represent processes, like actions or tasks and arrows to show how data moves between them. Data can be stored in places called data stores like files or databases and external entities represent where data comes from or goes to like users or other systems. DFDs help us understand how data is processed and stored in a system making it easier to identify problems or areas for improvement. This visual representation makes it easier for developers and stakeholders to understand the system structure. As a result DFDs support better planning, analysis and system development.

3.2.1 Level 0 Data Flow Diagram

A Level 0 Data Flow Diagram (DFD) provides a simple and high level overview of how data moves within a system. It shows the main system as a single process and illustrates how external entities interact with it. This diagram helps in understanding how information enters the system, how it is processed and how results are returned to the users without showing the internal processing details.

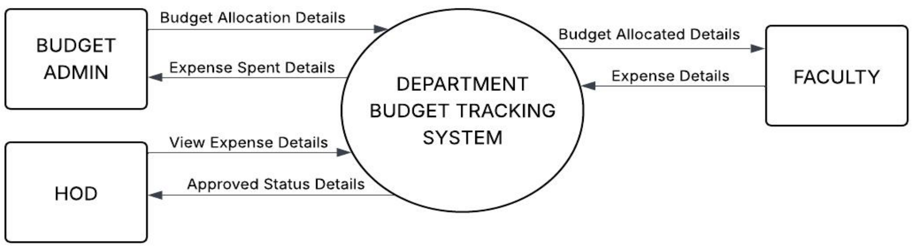


Figure 3.1 Level 0 Data Flow Diagram

The Figure 3.1 shows the Level 0 Data Flow Diagram of the Department Budget Tracking System, the main system process is represented as the Department Budget Tracking System. The system interacts with three external entities such as Budget Admin, Faculty and HOD. The Budget Admin provides budget allocation details to the system and receives information about the expenses spent. The Faculty submits expense details to the system and receives the allocated budget information. The HOD can view the expense details and receives the approved status details from the system. The figure 3.1 gives a clear overview of how data flows between users and the system in the Department Budget Tracking System.

3.2.2 Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram provides a more detailed representation of the system by breaking down the main process shown in Level 0 into smaller processes. It explains how data flows between different modules of the system, external entities and data stores. The Level 1 DFD helps in understanding the internal functioning of the system and how different operations are performed.

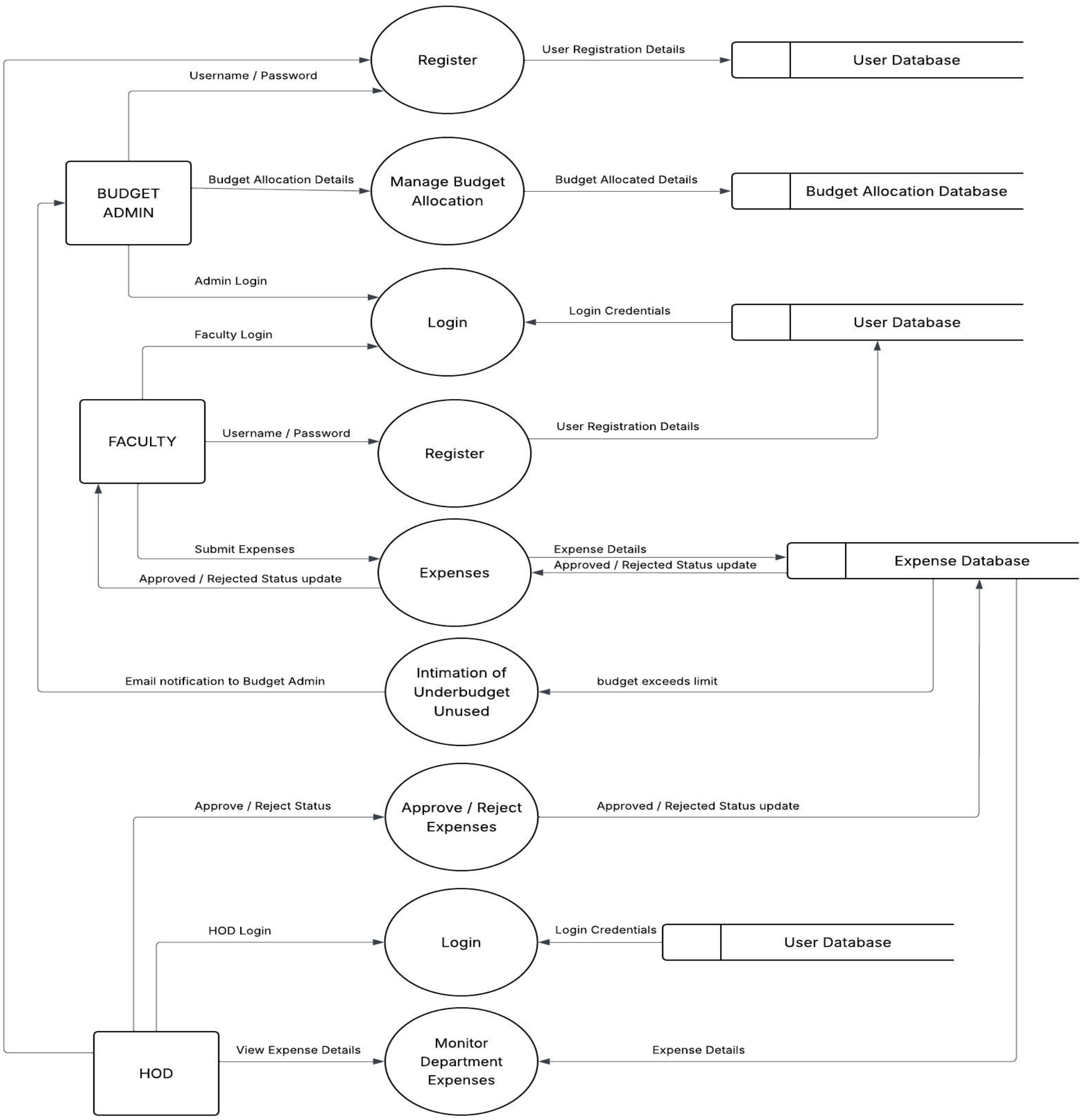


Figure 3.2 Level 1 Data Flow Diagram

Figure 3.2 shows the Level 1 Data Flow Diagram of the Department Budget Tracking System. The system includes three main external entities: Budget Admin, Faculty and HOD. The Budget Admin manages budget allocation and stores the details in the Budget Allocation Database. Faculty members log in and submit expense details related to departmental activities, which are stored in the Expense Database. The HOD reviews and approves or rejects the submitted expenses and monitors departmental spending. The system also generates notifications when the budget exceeds the defined limit and all user login details are maintained in the User Database.

3.3 USE CASE DIAGRAM

A Use Case Diagram is used to represent the interaction between users and the system. It shows the different functionalities of the system and how different users perform actions within the system. The diagram helps in understanding the roles of different users and the operations they can perform in the system. It provides a clear view of how users interact with the system to complete different tasks.

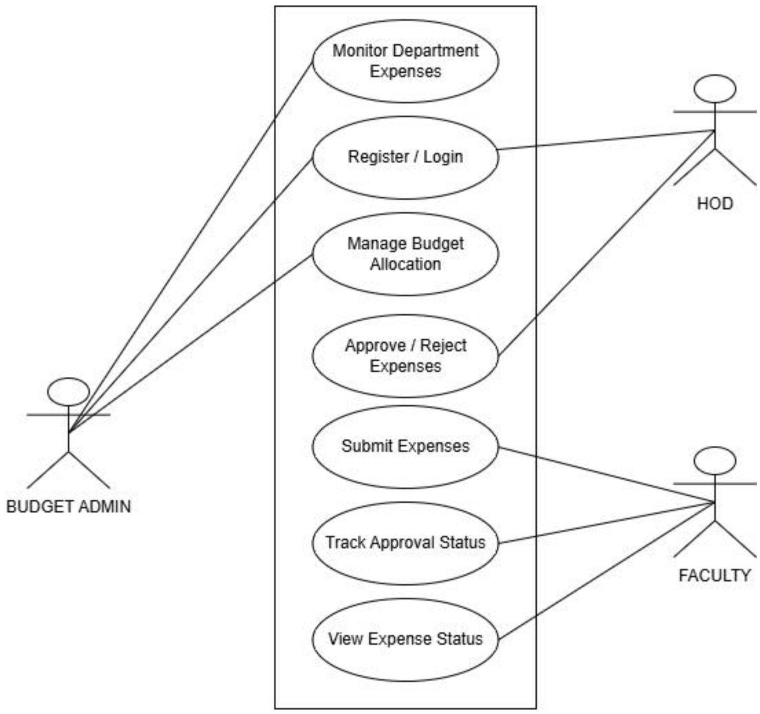
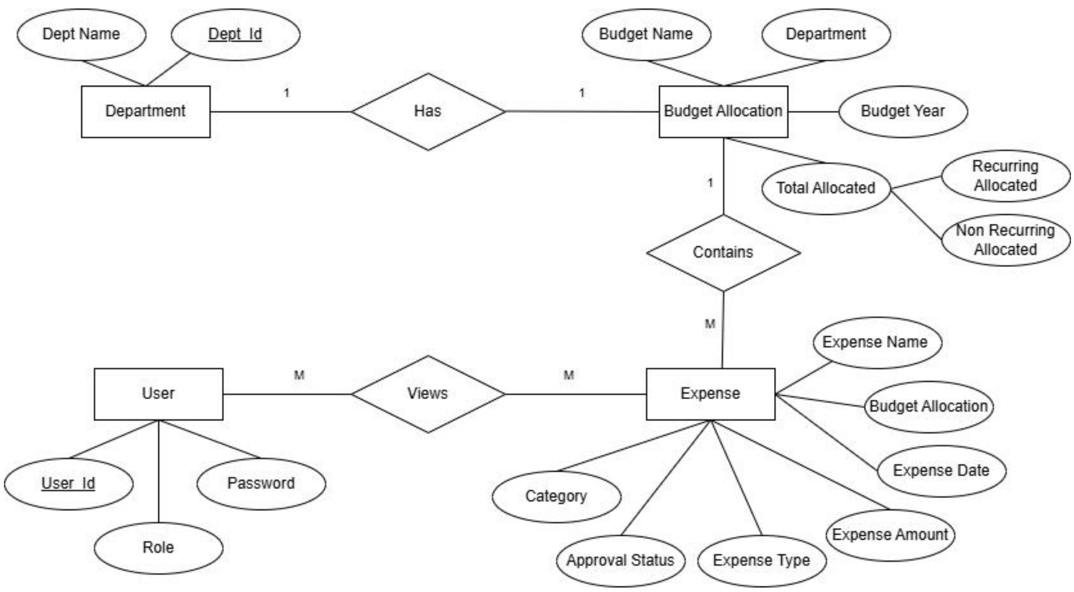


Figure 3.3 Use Case Diagram

The Figure 3.3 illustrates the Use Case Diagram of the Department Budget Tracking System. The diagram shows three main actors such as Budget Admin, Faculty and HOD who interact with the system to perform various operations.The Budget Admin plays an important role in managing the department budget. The Budget Admin can register and log in to the system manage the budget allocation and monitor the department expenses.The Faculty members interact with the system to submit expense details related to department activities. They can also track the approval status of their submitted expenses and view the final expense status after the approval process. The HOD mainly monitors the department's financial activities, approve or reject expenses submitted by faculty members. The HOD can log in to the system and view the expense details and approval status to ensure that the department budget is being used properly. The Figure 3.3 provides a clear representation of how different users interact with the Department Budget Tracking System and how the system supports budget management and expense monitoring within the department.

3.4 ER DIAGRAM

An Entity Relationship (ER) Diagram is used to represent the structure of a database system. It shows the entities, their attributes and the relationships between them. The ER diagram helps in understanding how data is organized and how different components of the system are connected.



3.4 ER Diagram

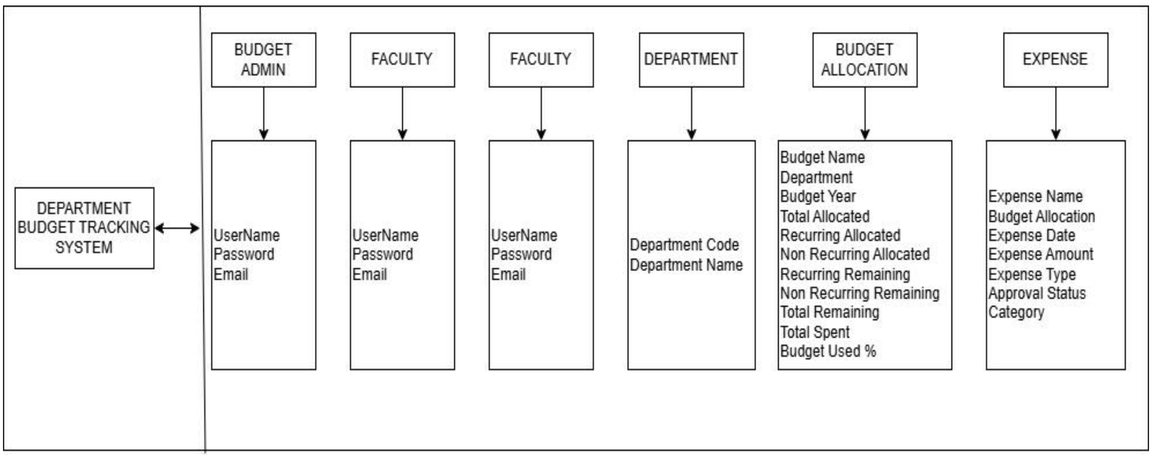
The Figure 3.4 illustrates the ER Diagram of the Department Budget Tracking System consists of four main entities such as Department, Budget Allocation, Expense, and

User. The Department entity contains attributes such as Department Name and Department ID. Each department can have a budget allocated for a particular year. The Budget Allocation entity stores information related to the department budget. It includes attributes such as Budget Name, Department, Budget Year, Total Allocated, Recurring Allocated and Non-Recurring Allocated. The relationship between Department and Budget Allocation indicates that each department has a budget allocation.

The Expense entity records the expenses made under the allocated budget. It contains attributes such as Expense Name, Budget Allocation, Expense Date, Expense Amount, Expense Type, Category and Approval Status. A budget allocation can contain multiple expenses. The User entity represents the system users such as Budget Admin, Faculty and HOD. It includes attributes such as User ID, Password and Role. Users can view and interact with the expense records in the system. The ER diagram clearly shows how the department budget is allocated, how expenses are recorded under each budget and how users interact with the system to manage and monitor the department budget.

3.5 DATABASE DESIGN

Database design refers to the process of organizing and structuring data so that it can be stored, managed and retrieved efficiently. A well designed database ensures that data is stored in a structured format and maintains consistency and accuracy.



3.5 Database Design Diagram of Budget Tracking System

The Figure 3.5 illustrates the Database Design Diagram consists of several main data tables such as Budget Admin, Faculty, Department, Budget Allocation and Expense.

Each table stores specific information related to the system functions. The Budget Admin and Faculty tables store user login details such as Username, Password and Email. These details are used for user authentication and system access. The Department table stores the information about departments, including Department Code and Department Name.

The figure 3.5 helps identify the department for which the budget is allocated. The Budget Allocation table stores the details related to the department budget. It includes fields such as Budget Name, Department, Budget Year, Total Allocated Amount, Recurring Allocated Amount, Non-Recurring Allocated Amount, Remaining Budget, Total Remaining Amount, Total Spent and Budget Used Percentage. These details help in tracking how much budget has been allocated and how much has been utilized.

The Expense table stores all the expense records submitted by faculty members. It includes attributes such as Expense Name, Budget Allocation, Expense Date, Expense Amount, Expense Type, Approval Status and Category. These records help monitor and manage the expenses under each allocated budget. The table 3.5 database design ensures that all system data is stored in an organized manner and helps the Department Budget Tracking System efficiently manage and track department budget usage and expenses.

3.6 TABLE DESIGN

Table Design refers to the process of organizing data into structured tables in a database. Each table consists of rows and columns where rows represent records and columns represent fields. Table design defines the field names, data types and relationships between different tables. A well-designed table helps store and retrieve data efficiently. It also ensures data accuracy, consistency and easy management of information in the system.

Table 3.1 Department Table

Field Name	Data Type	Description
Department Code	Text(10)	Code of the department
Department Name	Text(80)	Name of the department

Table 3.1 Department Table shows the fields used to store department details in the system.It includes details such as the department code and department name. These details help identify the department for which the budget allocation and expense tracking are

maintained. The department information is used as a reference while creating budget allocations and recording expenses in the Department Budget Tracking System.

Table 3.2 Budget Allocation Table

Field Name	**Data Type**	**Description**
Budget Name	Text(80)	Name of the budget.
Department	Lookup(Department)	Reference to the department for which the budget is allocated.
Budget Year	Picklist	Year for the budget allocation.
Total Allocated	Currency(18, 0)	Total budget amount allocated
Recurring Allocated	Currency(18, 0)	Budget allocated for recurring.
Non Recurring Allocated	Currency(18, 0)	Budget allocated for non-recurring.
Recurring Remaining	Currency	Calculated remaining amount from the recurring budget.
Non Recurring Remaining	Currency	Calculated remaining amount from the non-recurring budget.
Total Remaining	Currency	Remaining total budget amount after expenses are deducted.
Total Spent	Currency	Total amount spent.
Budget Used %	Percent	Percentage of the budget utilized.

Table 3.2 Budget Allocation Table stores the budget allocation details in the system, including budget name, department, budget year, total allocated amount and recurring and non-recurring allocations. It also contains calculated fields such as remaining budget, total spent and budget used percentage, which help track budget usage for department expenses.

Table 3.3 Expense Table

Field Name	**Data Type**	**Description**
Expense Name	Text(80)	Name of the expense
Budget Allocation	Master-Detail	Links the expense to the specific budget allocation.
Expense Date	Date	Specifies the date on which the expense occurred.

Expense Amount	Currency(18, 0)	Amount spent for the particular expense.
Expense Type	Picklist	Indicates whether the expense belongs to recurring or non-recurring type.
Approval Status	Picklist	Shows the approval state of the expense.
Category	Picklist	Represents the category.
Purpose	Long Text Area	Provides a detailed description.

Table 3.3 Expense Table shows the fields used to store expense details in the system. It includes details such as the expense name, budget allocation, expense date, expense amount, expense type, approval status, category and purpose. These details help record and track the expenses submitted by faculty members under the allocated department budget in the Department Budget Tracking System.

3.7 INPUT DESIGN

Input design is an important part of system development. It focuses on how data is entered into the system by the users. Proper input design ensures that the data entered into the system is accurate, complete and easy to understand. It helps reduce errors during data entry and improves the overall efficiency of the system. In the Department Budget Tracking System input is provided by different users such as Budget Admin and Faculty through various forms available in the system. These forms are designed in a simple and structured way so that users can easily enter the required details.

Login Form

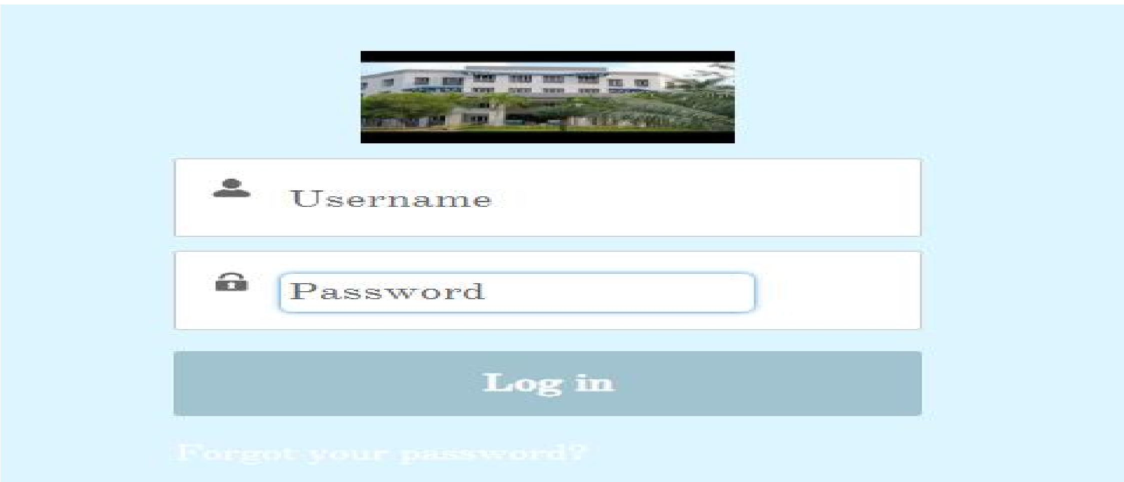


Figure 3.6 Login Form

Figure 3.6 shows the login form of the Department Budget Tracking System. This page allows authorized users such as Budget Admin, Faculty and HOD to access the system securely. The user must enter a valid username and password in the respective fields to log in. This login process ensures secure access to the system and protects the department budget information from unauthorized users.

Department Entry form

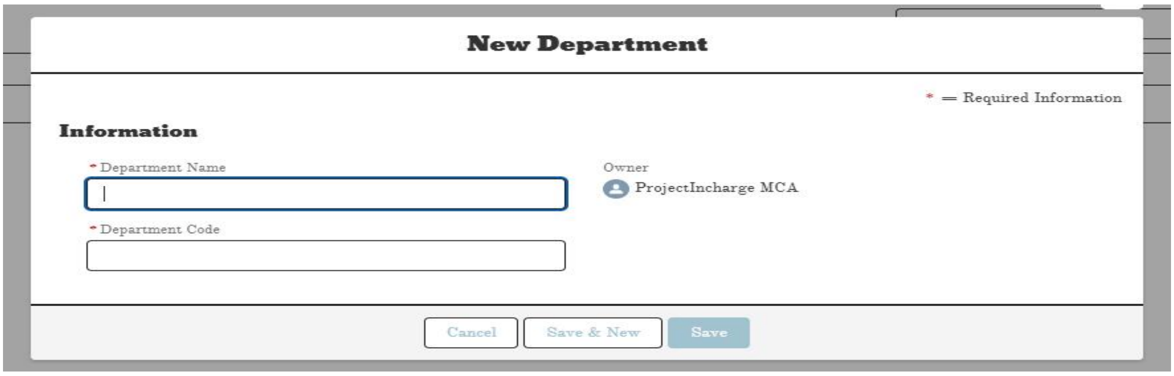


Figure 3.7 Department Entry Form

Figure 3.7 shows the Department Entry Form used in the Department Budget Tracking System. This form allows the Budget Admin to add new department details into the system. The admin must enter information such as the Department Name and Department Code. Once the required details are entered the admin can save the information and the department data will be stored in the system database for further budget allocation and expense tracking.

Budget Allocation Entry form

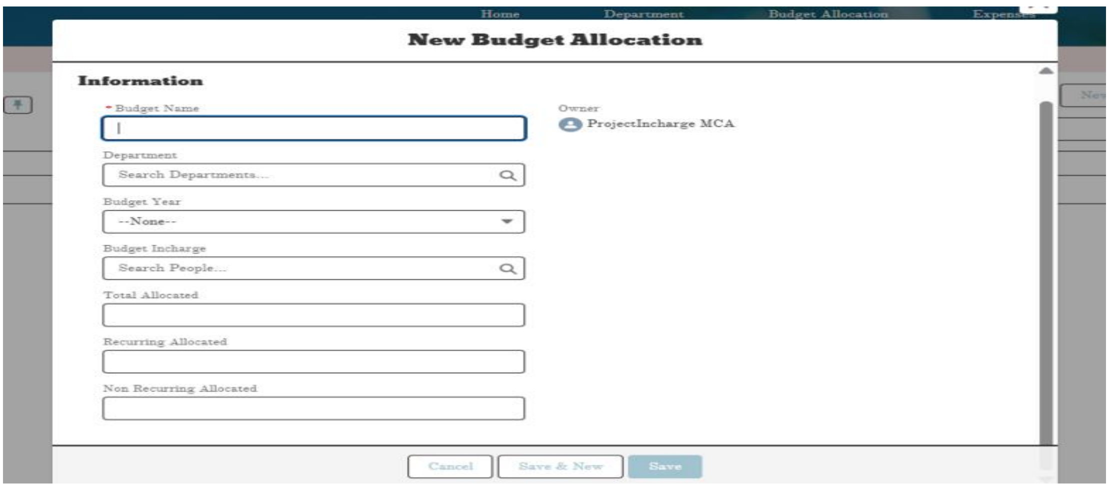


Figure 3.8 Budget Allocation Entry Form

Figure 3.8 shows the Budget Allocation Form used in the Department Budget Tracking System. This form allows the Budget Admin to allocate the budget for a specific department for a particular academic year. The admin enters details such as Budget Name, Department, Budget Year and Budget Incharge. The form also includes fields for Total Allocated Amount, Recurring Allocated Amount and Non-Recurring Allocated Amount. After entering all the required information, the admin can save the details and the budget allocation information will be stored in the system for managing and tracking department expenses.

Expense Entry Form

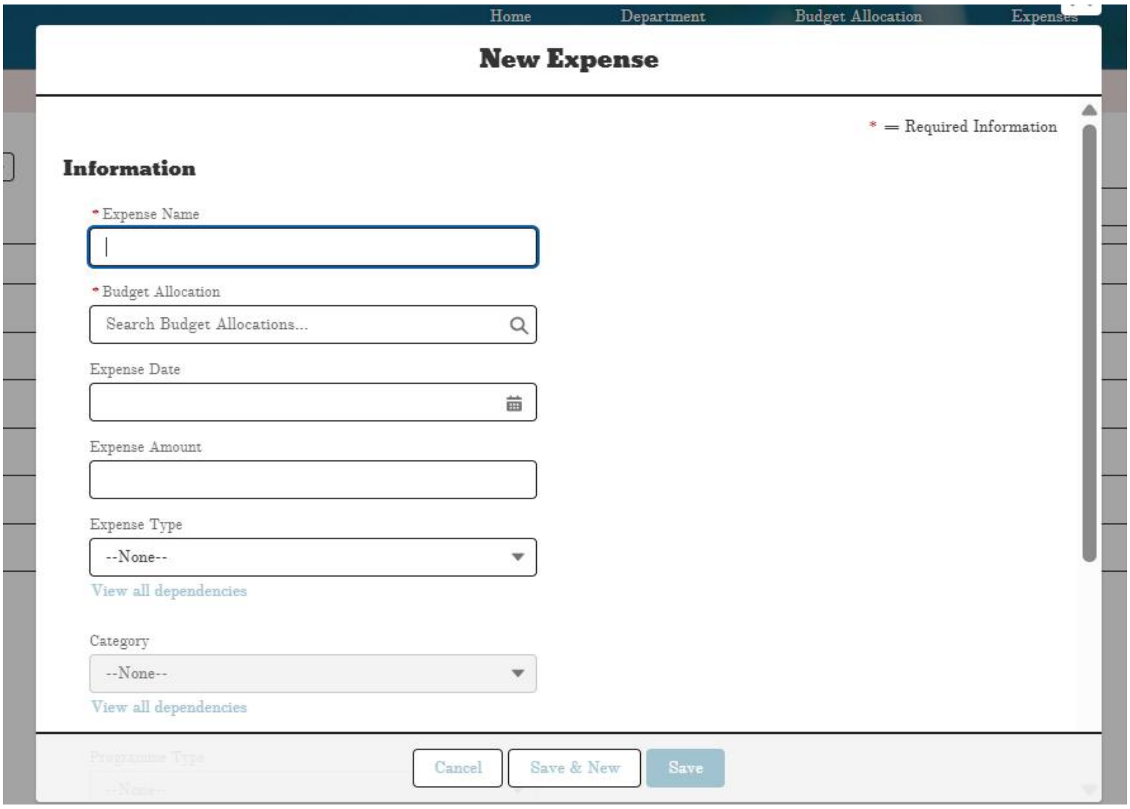


Figure 3.9 Expense Entry Form

Figure 3.9 shows the Expense Entry Form in the Department Budget Tracking System. This form allows faculty members to submit expense details related to department activities. The user enters information such as Expense Name, Budget Allocation, Expense Date, Expense Amount, Expense Type and Category. These details help in recording and tracking the expenses under the allocated department budget. Once the information is entered, the user can save the expense and the data will be stored in the Expense Database for approval and further monitoring.

3.8 OUTPUT DESIGN

Output design refers to the process of presenting the processed information to the users in a clear and understandable format. The main objective of output design is to provide meaningful and useful information that helps users understand the results generated by the system. A well designed output ensures that the information is accurate, organized and easy to interpret. In a system, outputs can be displayed in different forms such as reports, dashboards, tables, notifications or status updates. Proper output design helps users quickly analyze the data and make decisions based on the results provided by the system. In the Department Budget Tracking System, the outputs mainly include budget allocation details, expense records, approval or rejection status and expense monitoring reports. These outputs help users track the department budget usage and understand the financial activities within the system.

Expense Type Report

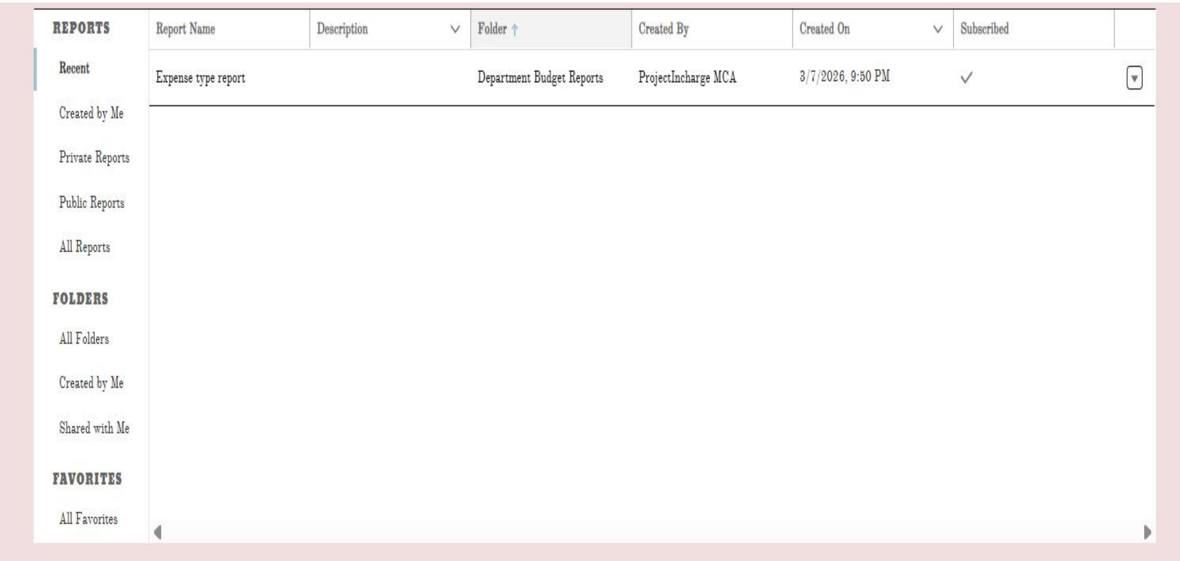
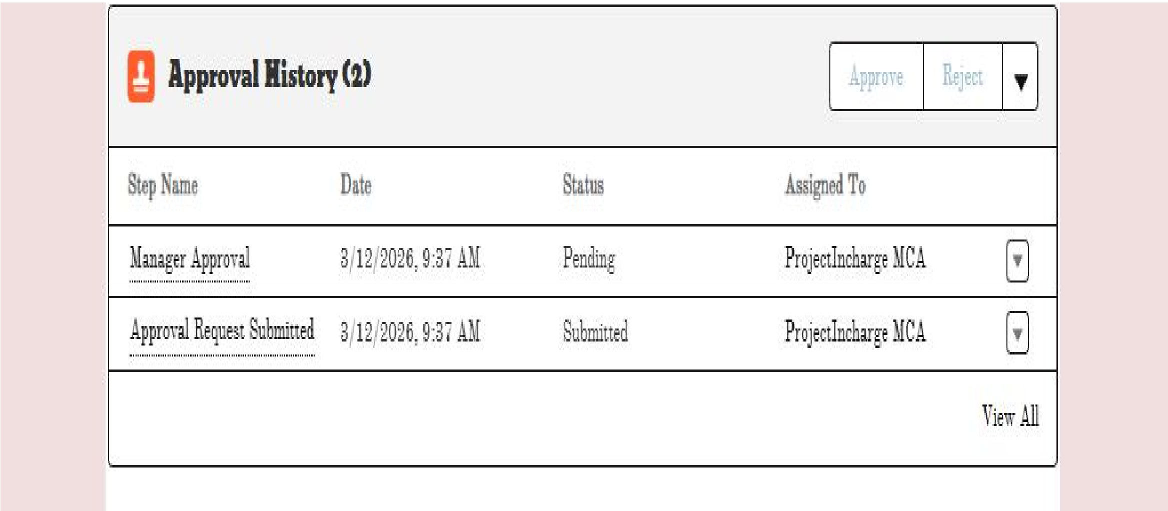


Figure 3.10 Expense Type Report

Figure 3.10 shows the Expense Type Report generated in the Department Budget Tracking System. This report provides a summary of the expenses recorded under different categories in the department. It helps users analyze how the allocated budget is being utilized for various activities. The report includes details such as the report name, folder, creator information, and creation date. By using this report, the Budget Admin and HOD can easily monitor department expenses and understand the distribution of spending across different expense types.

Expense Approval Request



3.11 Expense Approval Request

Figure 3.11 shows the Expense Approval Request in the Department Budget Tracking System. This screen displays the approval workflow of an expense submitted by the faculty member. It shows details such as the approval step name, date, current status and the person assigned to review the request. The HOD can review the request and either approve or reject the submitted expense. This feature helps manage the approval process and ensures that all expenses are properly verified before being finalized.

Budget Usage Alert Notification

3.12 Budget Usage Alert Notification

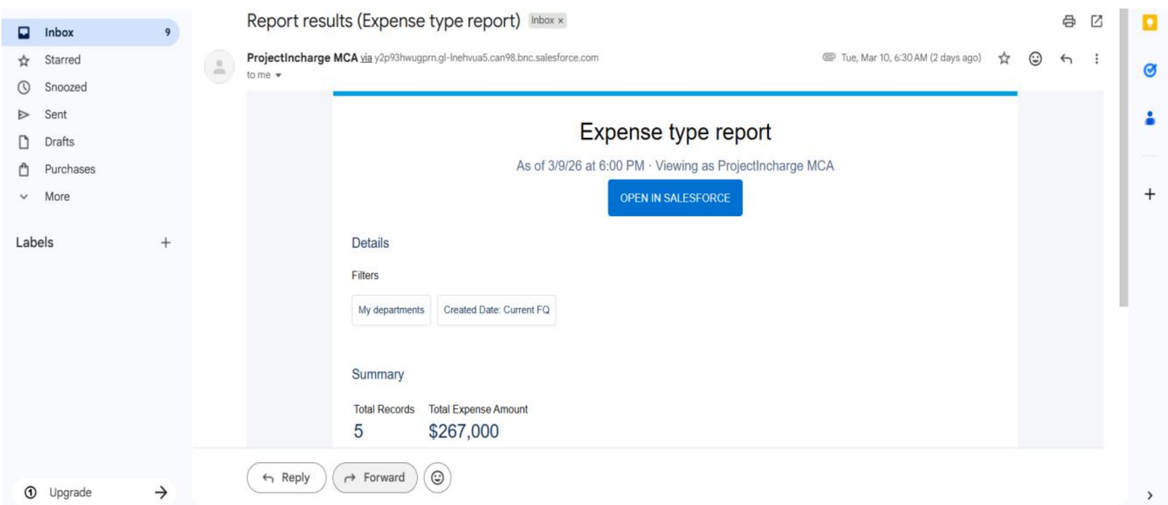
Figure 3.12 shows the Budget Usage Alert Notification generated by the Department Budget Tracking System. This notification is automatically sent to the Budget Admin when the department budget usage exceeds a predefined limit. In this case, the system indicates that the budget allocation for the MCA department has crossed 75% of its usage. The alert message includes details such as the department name, total allocated budget, remaining budget, and the percentage of budget used. This notification helps the Budget Admin monitor the budget utilization and take necessary action to control further expenses.

High Expense Approval Notification

3.13 High Expense Approval Notification

Figure 3.13 shows the High Expense Approval Notification generated by the Department Budget Tracking System. This notification is automatically sent when a submitted expense exceeds a predefined limit and requires approval from the Budget Admin. The message includes details such as the expense name and the expense amount. It informs the Budget Admin that a high value expense request has been submitted and requires approval or rejection. This feature helps ensure that large expenses are properly reviewed before being finalized in the system.

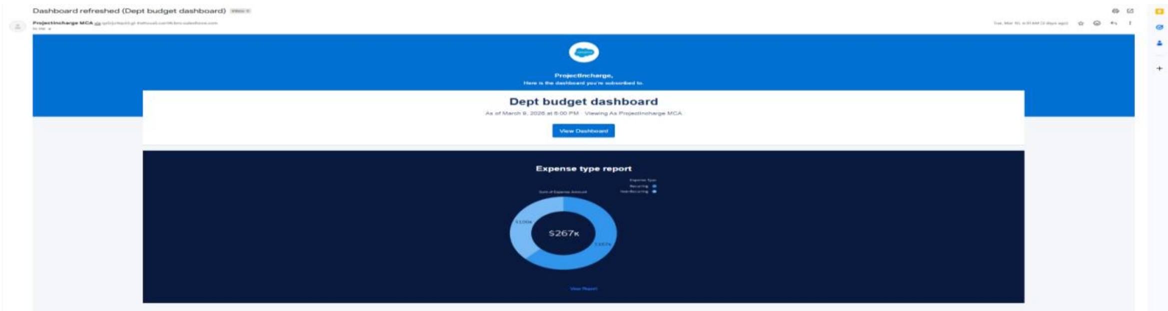
Monthly Expense Report Notification



3.14 Monthly Expense Report Notification

Figure 3.14 shows the Monthly Expense Report Notification generated by the Department Budget Tracking System. This email is automatically sent to the Budget Admin to provide a monthly summary of the department expense report. The report displays details such as filters used for the report, total number of expense records, and the total expense amount for the selected period. The email also includes a link to open the report directly in Salesforce for detailed viewing. This feature helps users regularly monitor and analyze the department's expense activities.

Dashboard Email Notification



3.15 Dashboard Email Notification

Figure 3.15 shows the Monthly Dashboard Email Notification generated by the Department Budget Tracking System. This email is automatically sent to the Budget Admin to provide a monthly summary of the department budget dashboard. The email contains a snapshot of the dashboard that displays the expense type report and the total expense amount.

SUMMARY

Chapter 3 describes the overall design of the Department Budget Tracking System and explains how the system components work together. It includes the description of different modules that perform specific functions within the system. The chapter also presents Data Flow Diagrams (Level 0 and Level 1) to show how data moves through the system and how users interact with it. In addition the system flow diagram illustrates the workflow of the application. The input and output design sections explain how data is entered into the system and how results are displayed. Finally, the ER diagram, database design and table design describe how the system data is organized and stored for effective budget management and expense tracking.

CHAPTER 4

IMPLEMENTATION

4.1 CODE DESCRIPTION

The Department Budget Tracking System is implemented using the Salesforce platform, which mainly uses configuration and automation tools instead of traditional programming code. The system uses Salesforce features such as Custom Objects, Flows, Approval Processes, Reports, Dashboards and Experience Cloud to manage department budgets and expenses. Salesforce Flows automate processes like sending email notifications when budget limits are exceeded. The Approval Process allows the HOD to approve or reject expenses submitted by faculty. Custom objects such as Department, Budget Allocation and Expense store the system data, while Experience Cloud provides the interface for users to access and manage the budget information.

4.2 STANDARDIZATION OF THE CODING

```
force-app > main > default > lwc > fetchsmartcomponent > JS fetchsmartcomponent.js > ...
import { LightningElement, api, wire } from 'lwc';
import getBudget from '@salesforce/apex/BudgetController.getBudget';
import getExpenses from '@salesforce/apex/BudgetController.getExpenses';
Qodo: Test this class
export default class BudgetDashboard extends LightningElement {
@api recordId;
budget;
expenses=[];
Qodo: Test this method
@wire(getBudget,{recordId:'$recordId'})
wiredBudget({data}){
if(data){
this.budget=data;
}
}
Qodo: Test this method
@wire(getExpenses,{budgetId:'$recordId'})
wiredExpenses({data}){
if(data){
this.expenses=data;
}
}
}
```

Figure 4.1 Standardized Code

Figure 4.1 shows the standardized code used in the Department Budget Tracking System developed using Salesforce Lightning Web Components (LWC). The LightningElement class is extended to create a reusable component that interacts with Salesforce records. The @api recordId property allows the component to access the current record ID from the Salesforce page. The @wire decorators are used to automatically retrieve budget and expense data from Apex controller methods such as getBudget and getExpenses.

Several coding standards are followed while developing this component. Meaningful naming conventions are used for classes, variables, and methods to improve clarity and understanding of the code. Component-based architecture is used so that the Lightning Web Component handles the user interface while the Apex controller manages the server side logic. The use of Salesforce decorators such as @api and @wire ensures proper data binding and secure communication between the frontend component and backend services. The code also follows consistent indentation and structured formatting, which improves readability and maintainability.

The modular design principles are applied so that each function performs a specific task such as retrieving budget data or fetching expense records. This separation of responsibilities makes the system easier to update and debug. Proper error prevention practices such as checking if data is available before assigning values are also followed to avoid runtime issues.

4.3 ERROR HANDLING

```
force-app > main > default > lwc > fetchsmartcomponent > JS fetchsmartcomponent.js > ...
import { LightningElement, api, wire } from 'lwc';
import getBudget from '@salesforce/apex/BudgetController.getBudget';
import getExpenses from '@salesforce/apex/BudgetController.getExpenses';
Qodo: Test this class
export default class BudgetDashboard extends LightningElement {
@api recordId;
budget;
expenses=[];
Qodo: Test this method
@wire(getBudget,{recordId:'$recordId'})
wiredBudget({data}){
if(data){
this.budget=data;
}
}
Qodo: Test this method
@wire(getExpenses,{budgetId:'$recordId'})
wiredExpenses({data}){
if(data){
this.expenses=data;
}
}
}
```

Figure 4.2 Standardized Error

Figure 4.2 shows the standardized error handling used in the Department Budget Tracking System developed using Salesforce Lightning Web Components (LWC). The code retrieves budget and expense details from the Salesforce database using Apex controller methods. The @wire decorators are used to automatically fetch data based on the current record ID. Proper conditional checks are included to ensure that data is processed only when valid results are returned. This structured approach helps prevent system errors, ensures smooth data retrieval and improves the reliability and stability of the application.

Several error handling and coding practices are followed in this implementation. Conditional validation is used to verify that returned data is available before assigning values to variables. This prevents runtime errors when the system receives null or empty responses. The use of separate methods for retrieving budget and expense data helps isolate errors and simplifies debugging. The code also follows modular design principles, where each function performs a specific task, making the system easier to maintain and update.

Proper naming conventions and structured formatting are used to improve code readability and maintain consistency. The component interacts with the Apex controller securely, ensuring that only authorized data is accessed from the Salesforce database. The use of decorators such as @api and @wire supports controlled data flow between the frontend and backend. The system also includes basic validation checks to ensure that incorrect or incomplete data is not processed. These practices help improve application stability, reduce system errors and ensure reliable data handling within the Department Budget Tracking System.

SUMMARY

Chapter 4 explains the implementation of the Department Budget Tracking System using the Salesforce platform. It describes how Salesforce features such as custom objects, flows, approval processes, reports and dashboards are used to develop the system. The chapter also discusses the coding standards followed to maintain consistency in the system components. In addition, error handling techniques are explained to ensure data accuracy and prevent invalid inputs. It also describes how Lightning Web Components and Apex controllers are used to retrieve and display data in the system. Furthermore, the chapter highlights how automation and structured development practices improve the efficiency and reliability of the application.

CHAPTER 5

TESTING AND RESULTS

Testing is an important phase in the development of the Department Budget Tracking System to ensure that the system functions correctly and meets the required objectives. It involves verifying that all modules such as department management, budget allocation, expense submission, approval workflow, email notifications and reporting are working properly without errors. Testing helps identify issues in the system and ensures that the system performs its intended functions effectively.

The testing is performed to check whether users can successfully create departments, allocate budgets, submit expenses, approve or reject expenses and monitor the budget through reports and dashboards. It also verifies that the expense data is stored correctly in the Salesforce database and that reports and dashboards display accurate information. Through proper testing the reliability and performance of the system are ensured before it is fully used. The testing techniques applied in this project include:

1. Unit Testing
2. Integration Testing
3. User Acceptance Testing

5.1 UNIT TESTING

Unit testing is performed to test individual modules of the system separately. Each module such as Department Management, Budget Allocation, Expense Submission, Approval Process and Email Notification is tested individually to ensure that it performs its intended function correctly. Unit testing helps identify errors in specific components before integrating them with other modules. It ensures that each module works independently without affecting other parts of the system. Unit testing also helps improve the reliability and quality of the system.

5.1.1 Department Management Module Testing

Figure 5.1 ensures that the administrator can successfully create a department by entering valid department details and the department record is stored correctly in the system.

Case 1:

Module: Department Management Module

Test Type: Department Creation Verification

Input: Department Code, Department Name

Output: Department Record Created Successfully

Test:

Input: Valid department details entered by administrator

Output: Department record saved in the Department object

Analysis: The department details are stored correctly in the system and appear in the department list.

Result: Pass

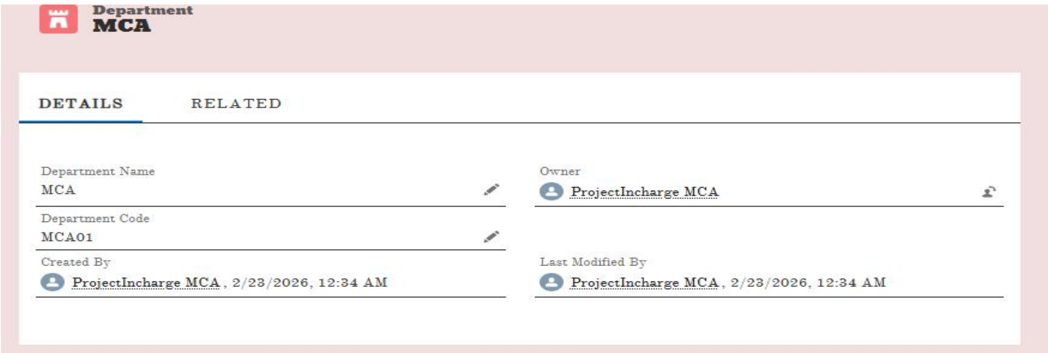


Figure 5.1 Department Management Module Testing

5.1.2 Budget Allocation Module Testing

Figure 5.2 ensures that the Budget Admin can successfully allocate a budget to a department by entering valid budget details, and the budget allocation record is stored correctly in the system.

Case 2:

Module: Budget Allocation Module

Test Type: Budget Allocation Verification

Input: Budget Name, Department, Budget Year, Allocated Amount

Output: Budget Allocation Record Created

Test:

Input: Valid budget details entered by Budget Admin

Output: Budget allocation stored in Budget Allocation object

Analysis: The system correctly stores the allocated budget details for the department.

Result: Pass

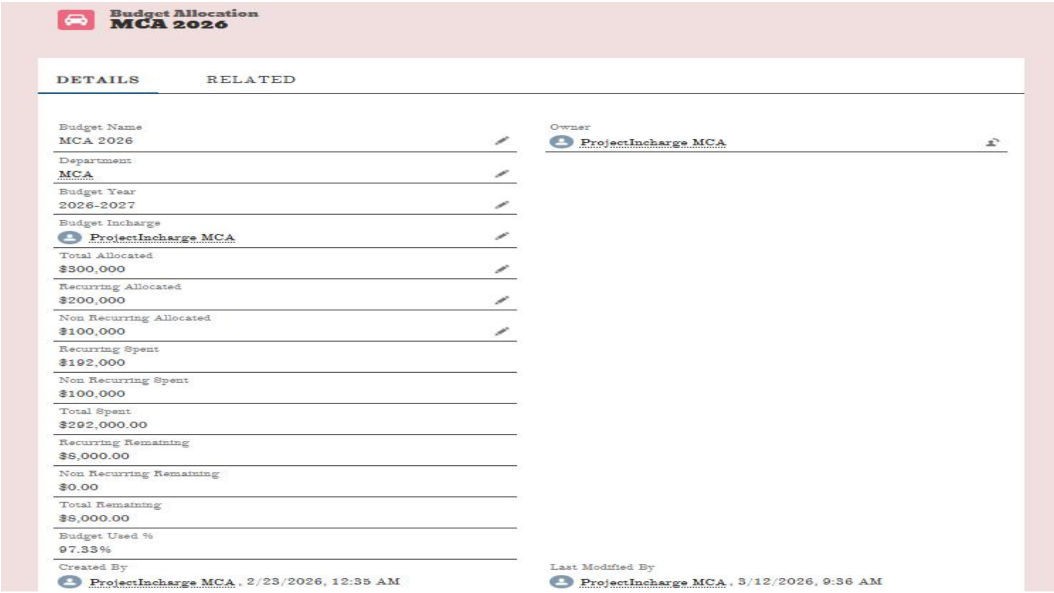


Figure 5.2 Budget Allocation Module Testing

5.1.3 Expense Submission Module Testing

Figure 5.3 ensures that faculty members can successfully submit expense details by entering valid information and the expense record is stored correctly in the system.

Case 3:

Module: Expense Submission Module

Test Type: Expense Entry Verification

Input: Expense Name, Expense Date, Expense Amount, Expense Type

Output: Expense Record Created

Test:

Input: Faculty submits valid expense details

Output: Expense record stored in Expense object

Analysis: The expense details are saved correctly and linked to the selected budget allocation.

Result: Pass

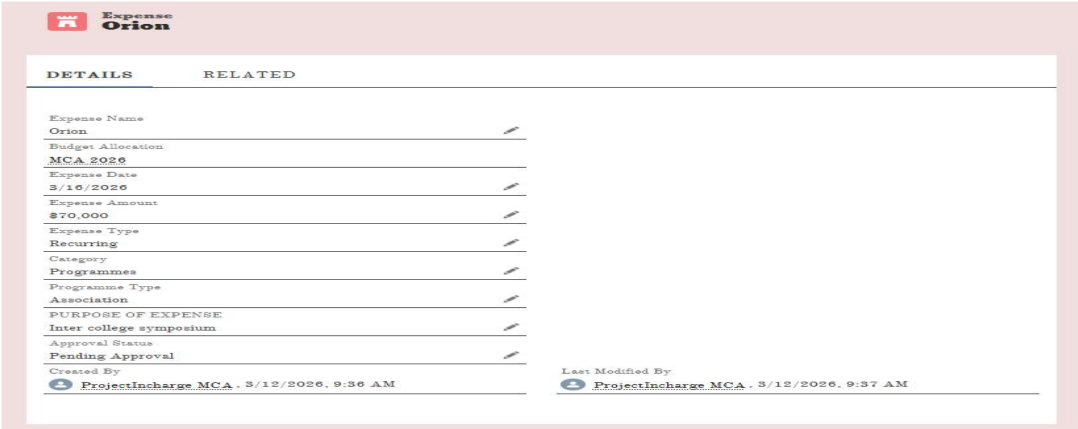


Figure 5.3 Expense Submission Module Testing

5.1.4 Approval Process Module Testing

Figure 5.4 ensures that the expense approval process works correctly when an expense is submitted for approval and the Budget Admin reviews the request.

Case 4:

Module: Approval Process Module

Test Type: Expense Approval Verification

Input: Expense submitted for approval

Output: Approval request sent to Budget Admin for review.

Test:

Input: Budget Admin clicks Approve option

Output: Expense status changes to Approved

Analysis: The approval process works correctly

Result: Pass

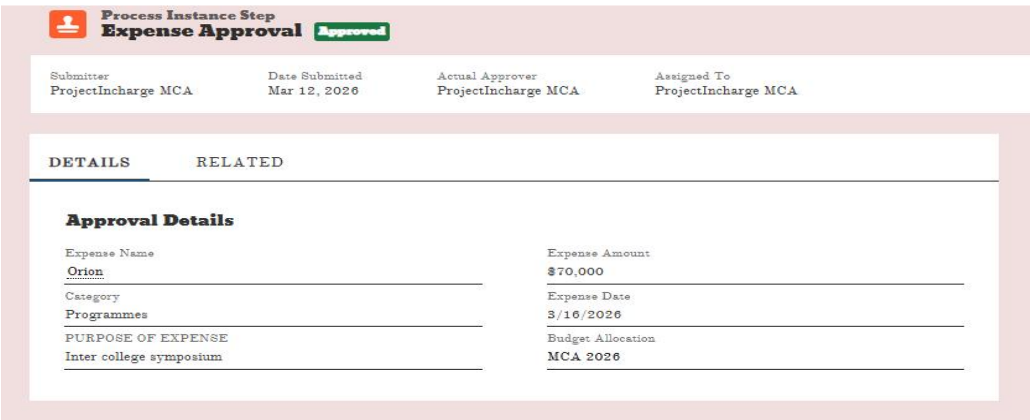


Figure 5.4 Approval Process Module Testing

5.2 INTEGRATION TESTING

Integration testing is used to verify that different modules of the system work together properly. In this project it ensures that once an expense is submitted by faculty, the system correctly sends the approval request to the Budget Admin and updates the budget details after approval. It confirms that the interaction between modules functions smoothly.

Integration Testing verifies that data flows correctly between modules such as department management, budget allocation, expense submission and approval workflow. Integration testing also ensures that the system correctly updates the remaining budget after expenses are recorded. It helps identify issues that occur when different modules interact with each other. It improves the overall reliability and coordination between all components of the system.

5.2.1 Department Module to Budget Allocation Module Testing

Figure 5.5 ensures that the department module and budget allocation module work together correctly when a new department is created and budget is allocated to it.

Case 1:

Modules: Department Module -> Budget Allocation Module

Test Type: Module Interaction Verification

Input: Administrator creates a new department

Output: Budget allocation can be successfully assigned to the created department.

Test:

Input: Budget Admin allocates budget to the department

Output: Budget allocation successfully linked with the department

Analysis: The system correctly associates budget allocation with the selected department.

Result: Pass

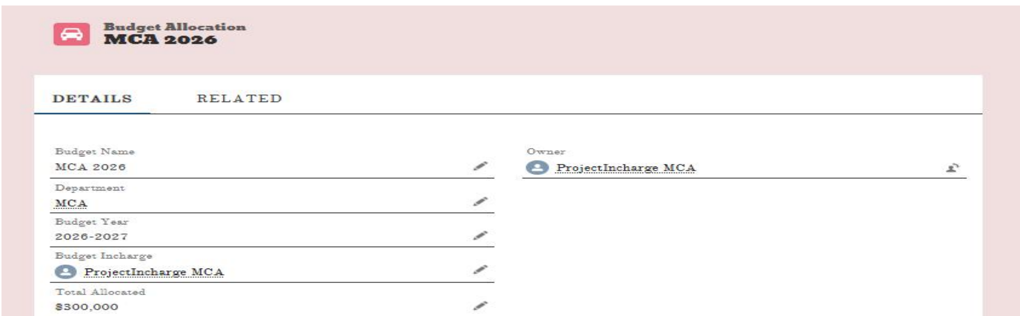


Figure 5.5 Department Module to Budget Allocation Module Testing

5.2.2 Budget Allocation Module to Expense Module Testing

Figure 5.6 and 5.7 ensures that the budget allocation module and expense module work together correctly when a faculty member submits an expense under the allocated budget.

Case 2:

Modules: Budget Allocation Module -> Expense Module

Test Type: Budget Usage Verification

Input: Faculty submits an expense

Output: Expense record is created and the budget spent amount is updated automatically.

Test:

Input: Expense linked with a budget allocation

Output: Budget spent value updated automatically

Analysis: The system updates the total spent and remaining budget correctly.

Result: Pass

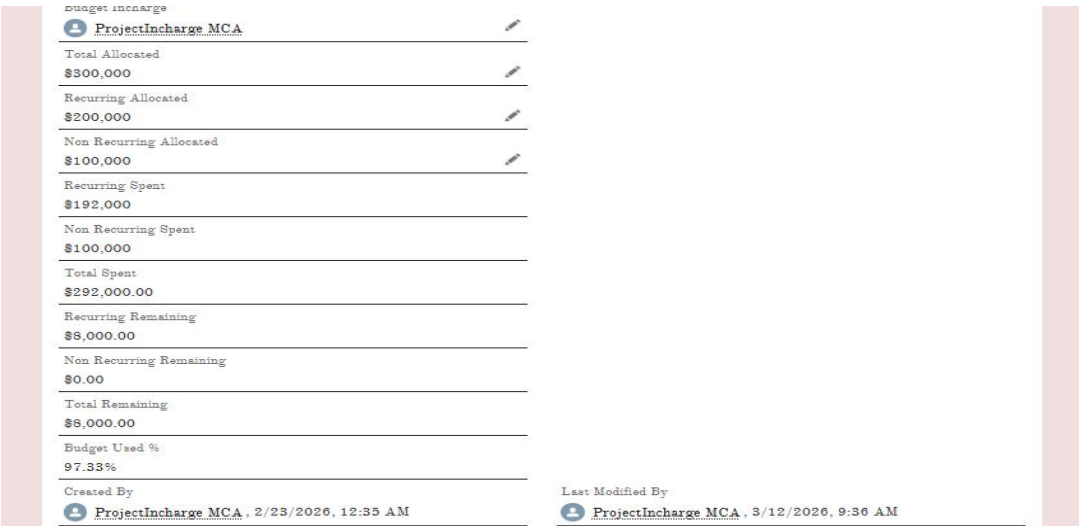


Figure 5.6 Budget Allocation Module to Expense Module Testing

Case 2

Modules : Budget Allocation Module -> Expense Module

Test Type : Budget Limit Validation

Input : Faculty submits an expense amount that exceeds the remaining allocated budget.

Output : System displays an error message and prevents saving the expense record.

Test

Input : Expense amount entered is greater than the available remaining budget.

Output : System shows the error message “Expense amount exceeds the available remaining budget” and the record is not saved.

Analysis : The system correctly validates the expense amount and restricts entries that exceed the remaining budget.

Result : Fail (Validation Triggered Successfully)

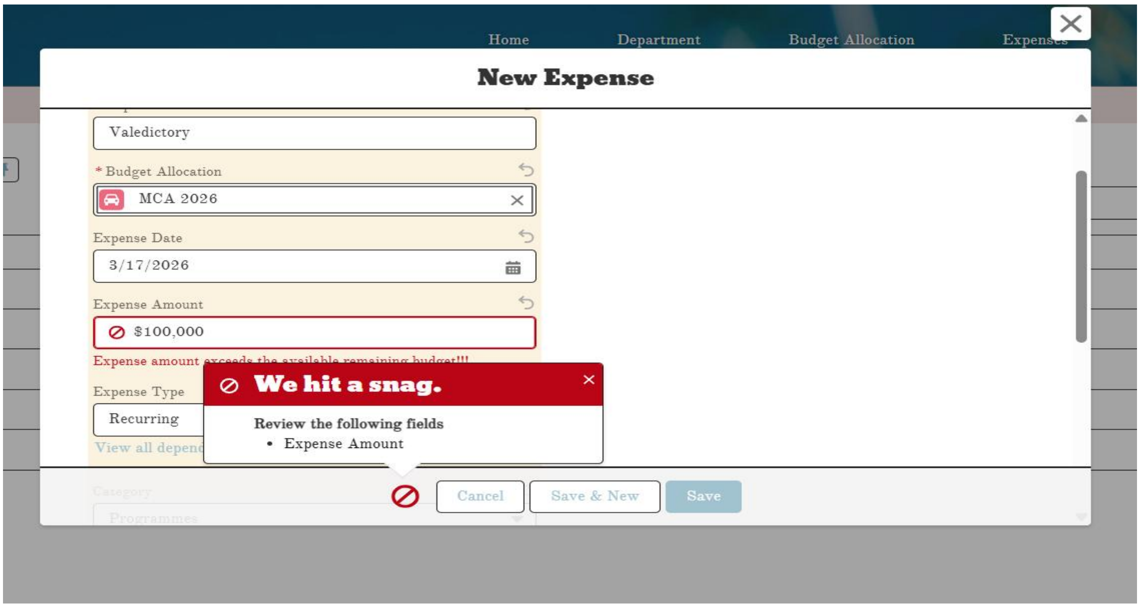


Figure 5.7 Budget Allocation Module to Expense Module Testing

5.2.3 Expense Module to Approval Process Module Testing

Figure 5.8 ensures that the expense module and approval process module work together correctly when a faculty member submits an expense and the approval request is sent to the Budget Admin.

Case 3:

Modules: Expense Module -> Approval Process Module

Test Type: Approval Trigger Verification

Input: Expense submitted by faculty

Output: Approval request is automatically sent to the Budget Admin for review.

Test:

Input: Budget Admin reviews the expense request

Output: Expense status changes to Approved or Rejected

Analysis: The approval process works correctly between the expense and approval modules.

Result: Pass

Figure 5.8 Expense Module to Approval Process Module Testing

5.3 USER ACCEPTANCE TESTING

User Acceptance Testing is conducted to ensure that the Department Budget Tracking System satisfies the requirements of the users. In this stage users such as Budget Admin, Faculty and HOD test the system to verify that budget allocation, expense submission and approval processes work correctly. This testing confirms that the system is user-friendly and ready for real time usage.

5.3.1 Budget Admin Testing

Figure 5.9 ensures that the Budget Admin can successfully create and manage department budget allocations by entering valid budget details.

Case 1
User : Budget Admin
Module : Budget Allocation Module
Test Type : Budget Allocation Verification
Test
Input : Budget Admin enters budget name, department, budget year and allocated amount
Output : Budget allocation record created successfully
Analysis : The Budget Admin is able to create and manage department budget allocations successfully.
Result : Pass

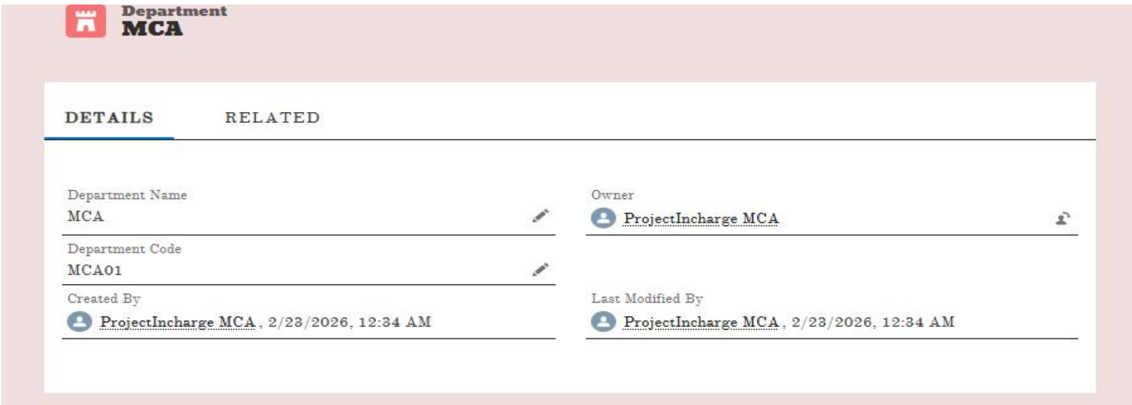


Figure 5.9 Budget Admin Testing

5.3.2 Faculty Testing

Figure 5.10 ensures that faculty members can successfully submit expense details by entering valid information, and the expense record is stored correctly in the system.

Case 2

User : Faculty

Module : Expense Submission Module

Test Type : Expense Entry Verification

Test

Input : Faculty enters expense name, expense date, amount and submits the expense form

Output : Expense record created successfully

Analysis : Faculty members are able to submit expense details but cannot modify budget allocation.

Result : Pass

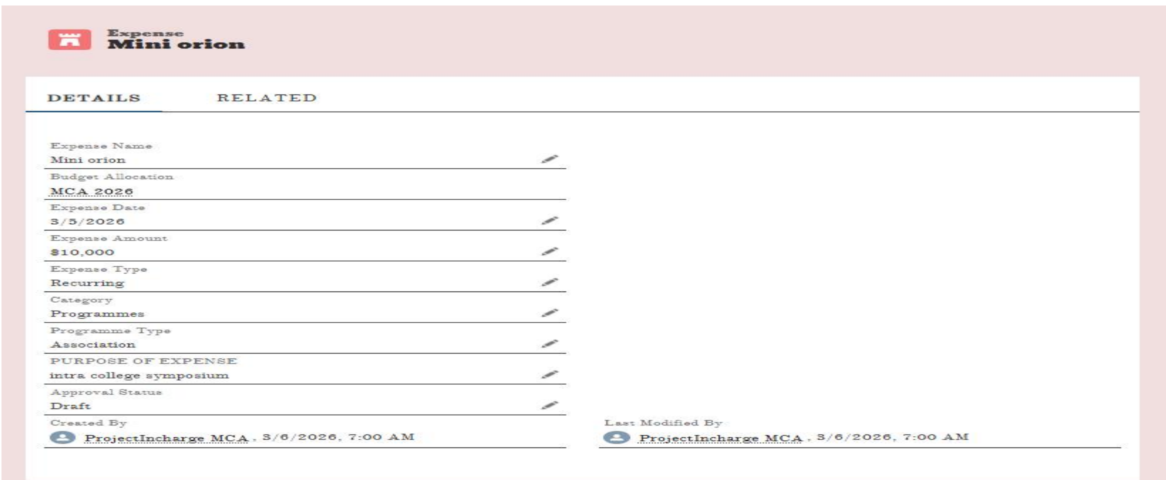


Figure 5.10 Faculty Testing

5.3.3 HOD Testing

Figure 5.11 ensures that the HOD can successfully review and approve or reject the expense requests submitted by faculty members.

Case 3

User : HOD

Module : Approval Process Module

Test Type : Expense Approval Verification

Test

Input : HOD reviews the submitted expense request

Output : Expense status changes to Approved or Rejected

Analysis : HOD is able to approve or reject expenses submitted by faculty members successfully.

Result : Pass

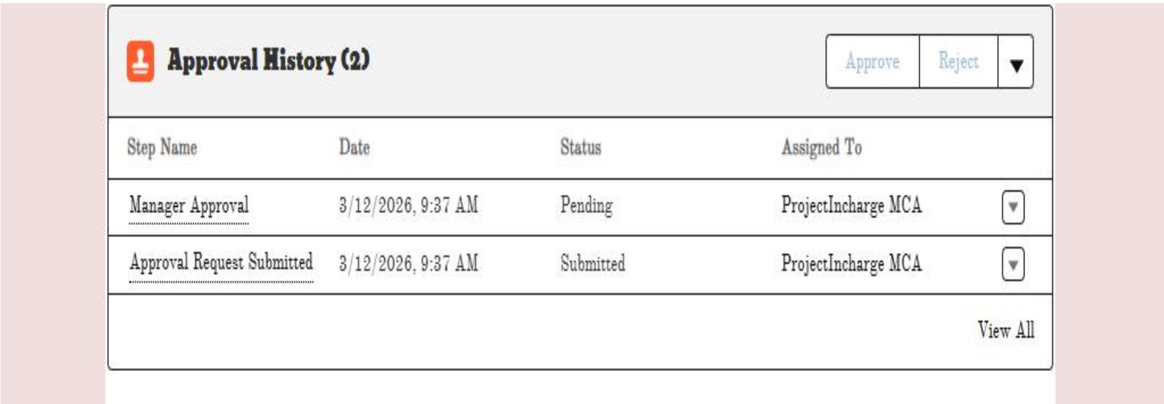


Figure 5.11 HOD Testing

SUMMARY

Chapter 5 explained the testing process carried out for the Department Budget Tracking System to ensure that the system functions correctly and meets the user requirements. Different types of testing such as module testing, integration testing and user acceptance testing were performed to verify the system functionality. Test cases were conducted for modules like department management, budget allocation, expense submission and approval process to ensure that each component works properly. The results show that the system performs as expected and supports accurate budget management and expense monitoring.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The Department Budget Tracking System developed using the Salesforce platform provides an efficient and automated solution for managing department budgets and expenses. The system simplifies the processes of budget allocation, expense submission and approval using Salesforce features such as Experience Cloud, Flows, Approval Processes, Reports and Dashboards. The Budget Admin manages budget allocations, faculty members submit expenses and the HOD reviews and approves them through a structured workflow. Once approved the system automatically updates the budget details and records the expense information. The system provides reports and dashboards that help monitor expenses, analyze budget usage and support better decision making. Automated calculations help track remaining budget and total spending accurately. Email notifications and alerts ensure that users are informed about expense approvals and budget limits. Overall the system improves transparency, efficiency and provides a reliable platform for effective budget management within the department.

6.2 FUTURE ENHANCEMENT

1. Automated Budget Alerts can enhance the system can be enhanced to send automatic notifications when the budget usage reaches a certain limit to help administrators control expenses.
2. Advanced Reporting Features is about the additional reporting and filtering options can be added to provide detailed analysis of department expenses and budget utilization.
3. Mobile Friendly Access can enhance the system can be improved by developing mobile access so that users can easily manage budgets and expenses using smartphones or tablets.

APPENDICES

A. SAMPLE CODING

expenseForm.html

```
<template>
<lightning-card title="New Expense">
<div class="slds-p-around_medium">
<lightning-record-edit-form
object-api-name="Expense__c"
onsuccess={handleSuccess}>
<lightning-messages></lightning-messages>
<!-- Expense Name -->
<lightning-input-field
field-name="Expense_Name__c">
</lightning-input-field>
<!-- Budget Allocation -->
<lightning-input-field
field-name="Budget_Allocation__c">
</lightning-input-field>
<!-- Expense Date -->
<lightning-input-field
field-name="Expense_Date__c">
</lightning-input-field>
<!-- Expense Amount -->
<lightning-input-field
field-name="Expense_Amount__c">
</lightning-input-field>
<!-- Expense Type -->
<lightning-input-field
```

```
field-name="Expense_Type__c">
</lightning-input-field>
<!-- Category -->
<lightning-input-field
field-name="Category__c">
</lightning-input-field>
<!-- Purpose -->
<lightning-input-field
field-name="Purpose__c">
</lightning-input-field>
<!-- Approval Status -->
<lightning-input-field
field-name="Approval_Status__c">
</lightning-input-field>
<div class="slds-m-top_medium">
<lightning-button
type="submit"
label="Save"
variant="brand">
</lightning-button>
<lightning-button
label="Cancel"
class="slds-m-left_small">
</lightning-button>
</div>
</lightning-record-edit-form>
</div>
</lightning-card>
</template>
```

expenseForm.js

```
import { LightningElement } from 'lwc';
import { ShowToastEvent } from 'lightning/platformShowToastEvent';
```

```
export default class ExpenseForm extends LightningElement {
handleSuccess() {
const event = new ShowToastEvent({
title: 'Success',
message: 'Expense created successfully',
variant: 'success'
});
this.dispatchEvent(event);
}
}
```

expenseForm.js-meta.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<LightningComponentBundle xmlns="http://soap.sforce.com/2006/04/metadata">
<apiVersion>60.0</apiVersion>
<isExposed>true</isExposed>
<targets>
<target>lightning__AppPage</target>
<target>lightning__RecordPage</target>
<target>lightning__HomePage</target>
</targets>
</LightningComponentBundle>
```

BudgetController.cls

```
public with sharing class BudgetController {
@AuraEnabled(cacheable=true)
public static Budget_Allocation__c getBudget(Id recordId) {
return [
SELECT Name,
Department__r.Name,
Budget_Year__c,
Total_Allocated__c,
```

```
Recurring_Allocated__c,
Non_Recurring_Allocated__c,
Recurring_Remaining__c,
Non_Recurring_Remaining__c,
Total_Remaining__c,
Budget_Used__c
FROM Budget_Allocation__c
WHERE Id = :recordId
];
}
@AuraEnabled(cacheable=true)
public static List<Expense__c> getExpenses(Id budgetId){
return [
SELECT Id, Expense_Name__c
FROM Expense__c
WHERE Budget_Allocation__c = :budgetId
];
}
}
```

budgetDashboard.html

```
<template>
<lightning-card title="Budget Details">
<div class="slds-grid slds-wrap">
<div class="slds-col slds-size_1-of-2">
<p><b>Budget Name :</b> {budget.Name}</p>
<p><b>Department :</b> {budget.Department__r.Name}</p>
<p><b>Budget Year :</b> {budget.Budget_Year__c}</p>
</div>
<div class="slds-col slds-size_1-of-2">
<p><b>Total Allocated :</b> {budget.Total_Allocated__c}</p>
<p><b>Recurring Allocated :</b> {budget.Recurring_Allocated__c}</p>
<p><b>Non Recurring Allocated :</b> {budget.Non_Recurring_Allocated__c}</p>
```

```
</div>
</div>
</lightning-card>
<lightning-card title="Expense Summary">
<p>Recurring Spent : {recurringSpent}</p>
<p>Total Spent : {totalSpent}</p>
</lightning-card>
<lightning-card title="Remaining Budget">
<p>Recurring Remaining : {budget.Recurring_Remaining__c}</p>
<p>Total Remaining : {budget.Total_Remaining__c}</p>
</lightning-card>
<lightning-card title="Expenses">
<template for:each={expenses} for:item="exp">
<p key={exp.Id}>{exp.Expense_Name__c}</p>
</template>
</lightning-card>
</template>
```

budgetDashboard.js

```
import { LightningElement, api, wire } from 'lwc';
import getBudget from '@salesforce/apex/BudgetController.getBudget';
import getExpenses from '@salesforce/apex/BudgetController.getExpenses';
export default class BudgetDashboard extends LightningElement {
@api recordId;
budget;
expenses=[];
@wire(getBudget,{recordId:'$recordId'})
wiredBudget({data}){
if(data){
this.budget=data;
}
}
@wire(getExpenses,{budgetId:'$recordId'})
```

```
wiredExpenses({data}){
if(data){
this.expenses=data;
}
}
}
```

XML

```
<?xml version="1.0" encoding="UTF-8"?>
<LightningComponentBundle xmlns="http://soap.sforce.com/2006/04/metadata">
<apiVersion>60.0</apiVersion>
<isExposed>true</isExposed>
<targets>
<target>lightning__RecordPage</target>
</targets>
</LightningComponentBundle>
```

B. SCREENSHOTS

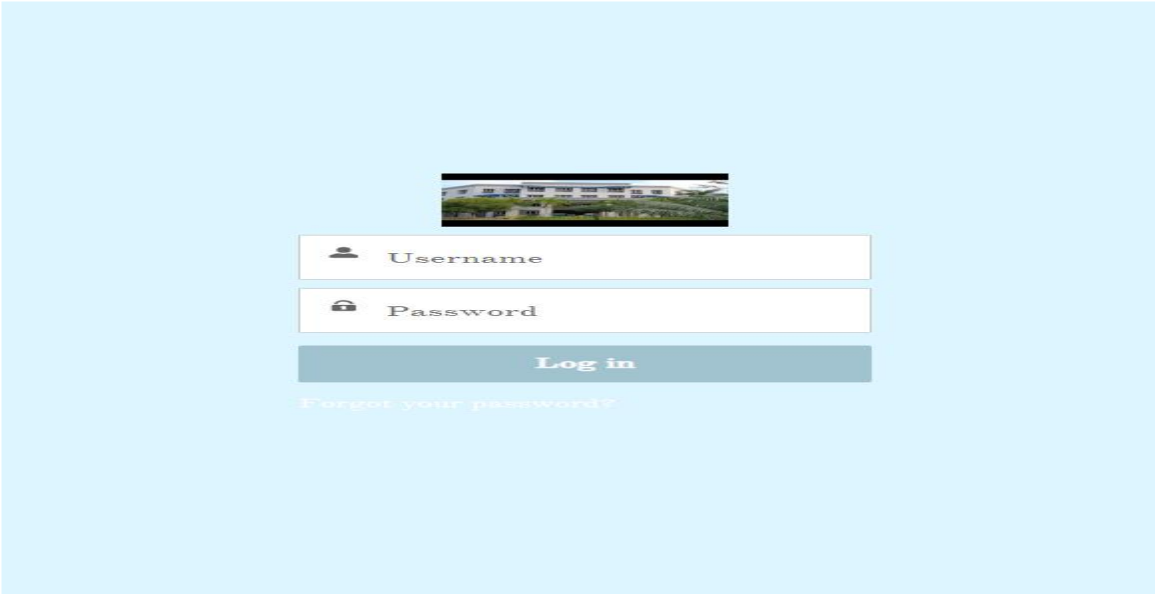


Figure B.1 Login Page

In Figure B.1 this is the login page of the Department Budget Tracking System, where users such as Budget Admin, Faculty, and HOD enter their username and password to access the system and manage departmental budget and expense activities.

Figure B.2 Home Page

In Figure B.2 this is the home page of the Department Budget Tracking System, where users can view the system title and access different modules of the system.

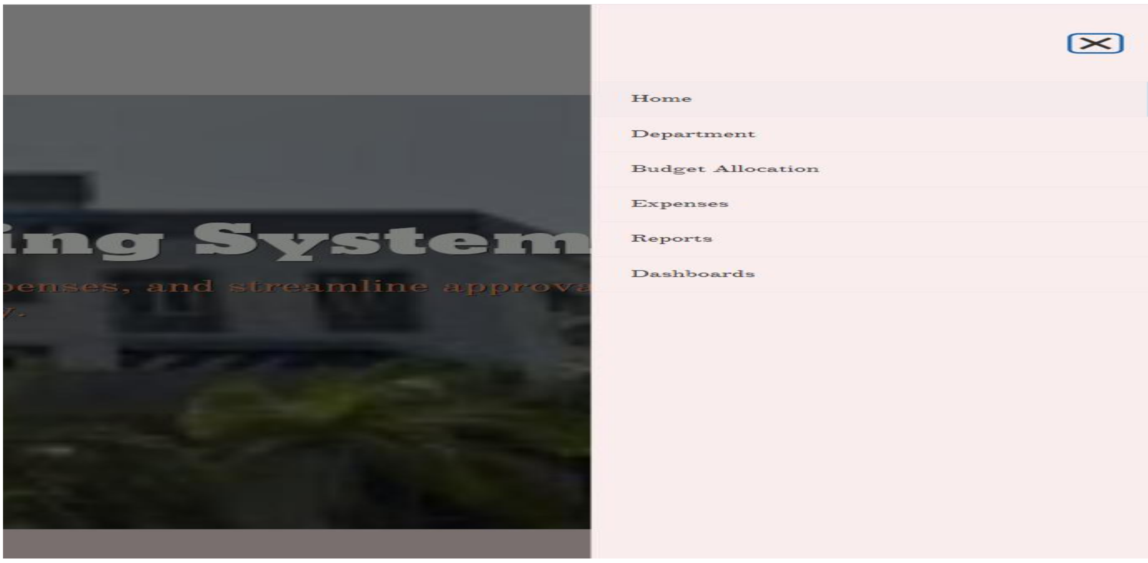


Figure B.3 Navigation Menu

In Figure B.3 this screen shows the navigation menu of the Department Budget Tracking System, where users can access different modules such as Home, Department, Budget Allocation, Expenses, Reports and Dashboards. This menu helps users easily navigate through the system and manage departmental budget and expense related activities.

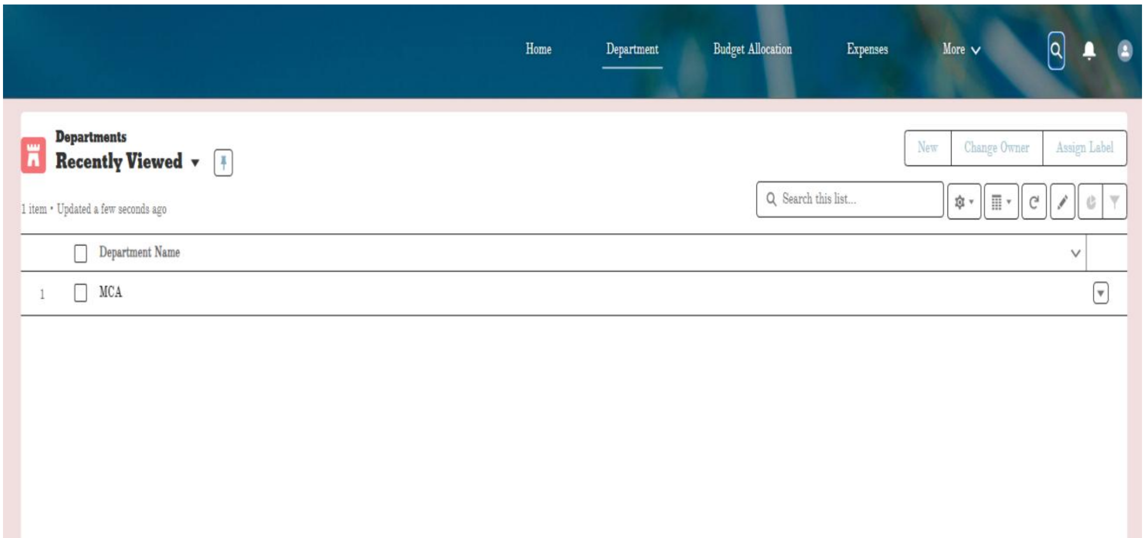


Figure B.4 Department Page

In Figure B.4 this screen shows the Department Management page of the Department Budget Tracking System. This page allows administrators to view and manage department records such as the department name. It helps in maintaining department details which are later used for budget allocation and expense tracking in the system.

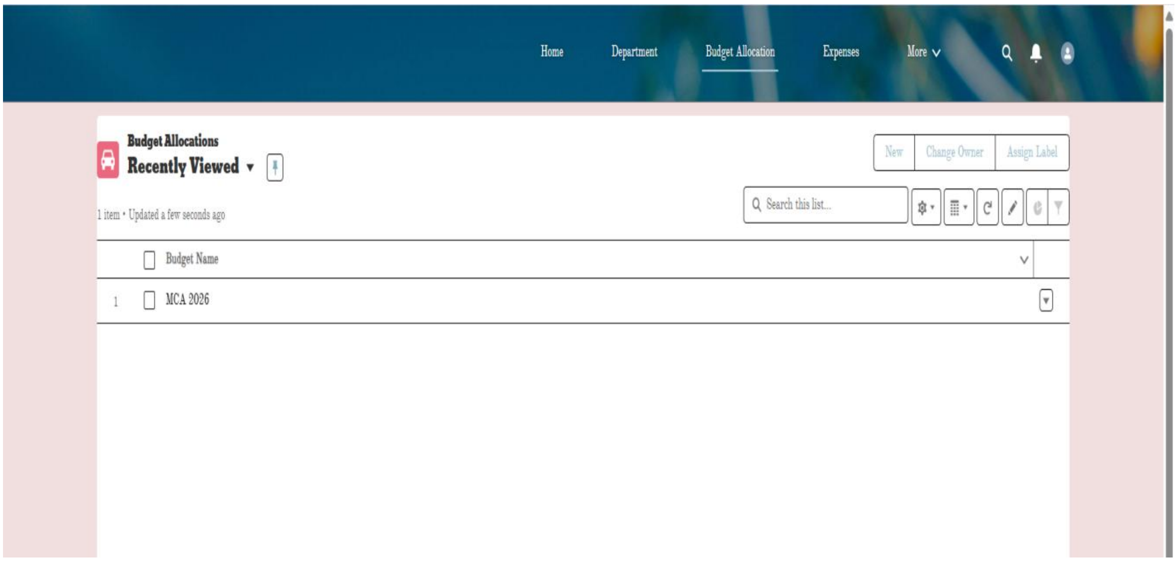


Figure B.5 Budget Allocation Page

In Figure B.5 this screen shows the Budget Allocation page of the Department Budget Tracking System. This page allows the Budget Admin to view and manage budget allocation records for the department. It displays details such as the budget name and helps administrators allocate and monitor budgets for departmental activities.

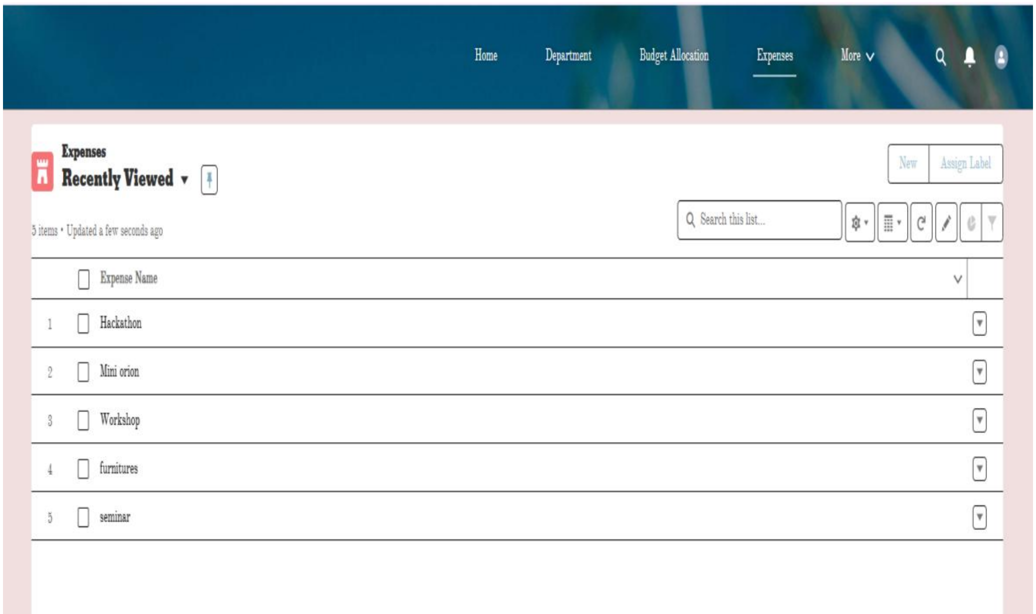


Figure B.6 Expense Page

In Figure B.6 this screen shows the Expense Management page of the Department Budget Tracking System. This page allows faculty members to view and manage expense records related to departmental activities such as workshops, seminars and events. It displays the list of expenses submitted and helps track departmental spending.

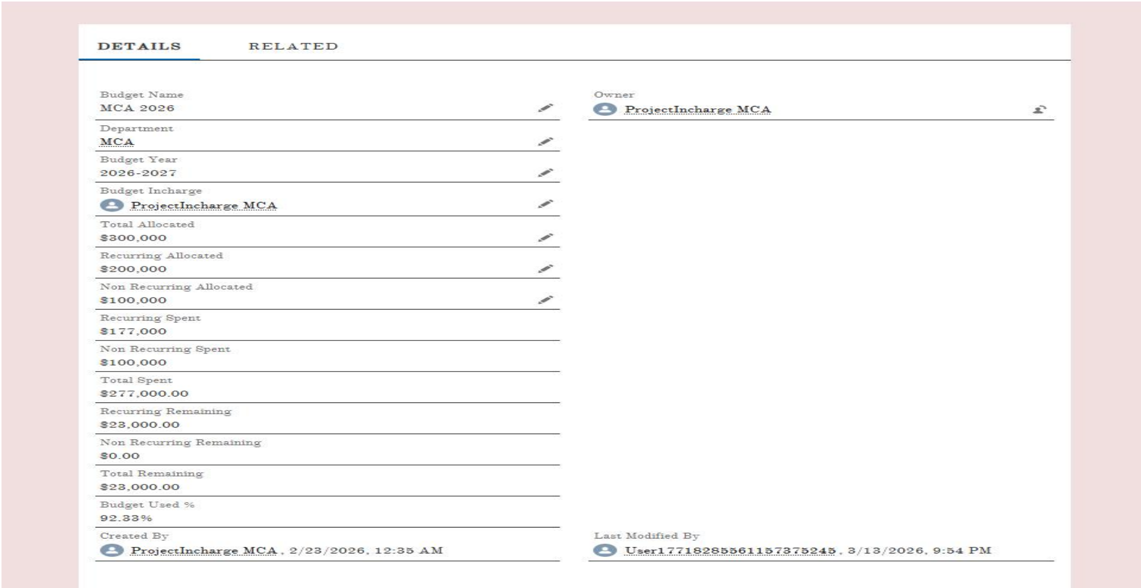


Figure B.7 Budget Allocation Details Page

In Figure B.7 this screen shows the Budget Allocation Details page of the Department Budget Tracking System. This page displays detailed information about the allocated budget for the department, including total allocated amount, recurring and non-recurring allocations, total spent, and remaining budget. It helps administrators monitor budget utilization and track departmental expenses effectively.

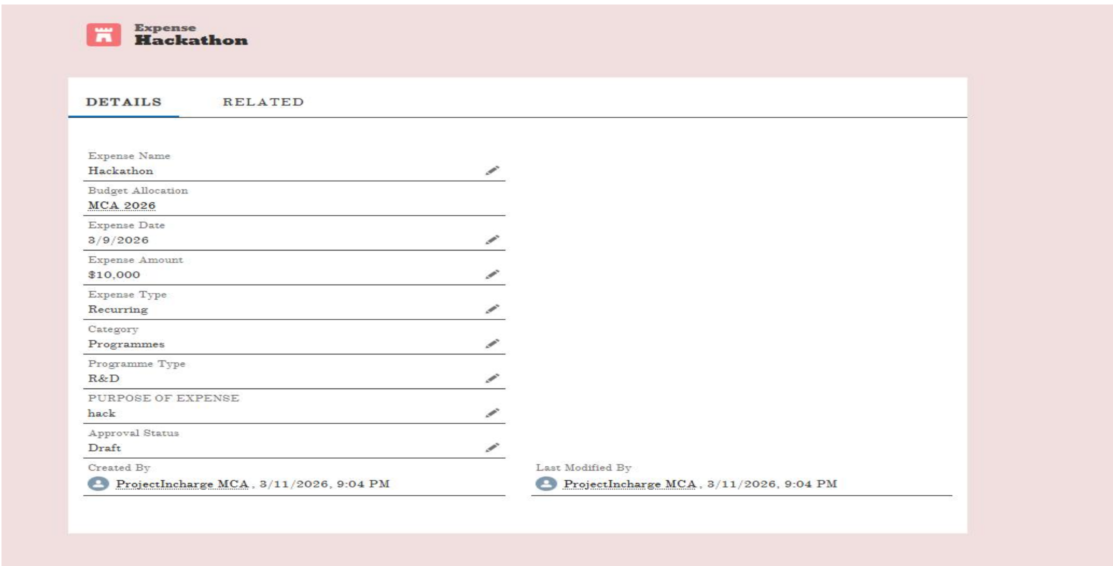


Figure B.8 Expense Details Page

In Figure B.8 this screen shows the Expense Details page of the Department Budget Tracking System. This page displays detailed information about a submitted expense, including the expense name, budget allocation, expense date, expense amount, expense type, category, and approval status. It helps administrators review and manage expense records related to departmental activities.

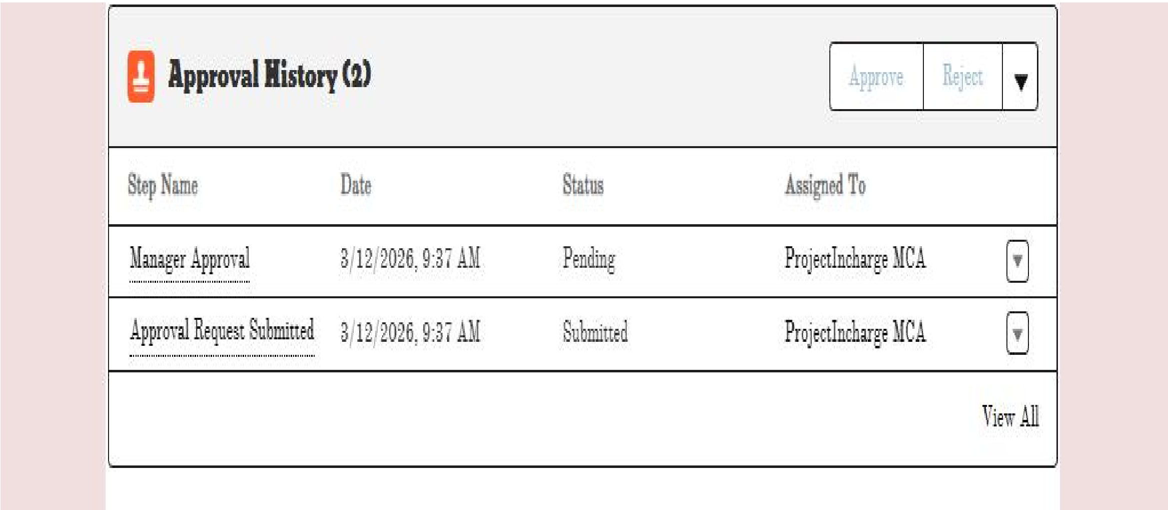


Figure B.9 Expense Approval Page

In Figure B.9 this screen shows the Expense Approval page of the Department Budget Tracking System. This page displays the approval history of an expense request, including the step name, date, status and assigned user.

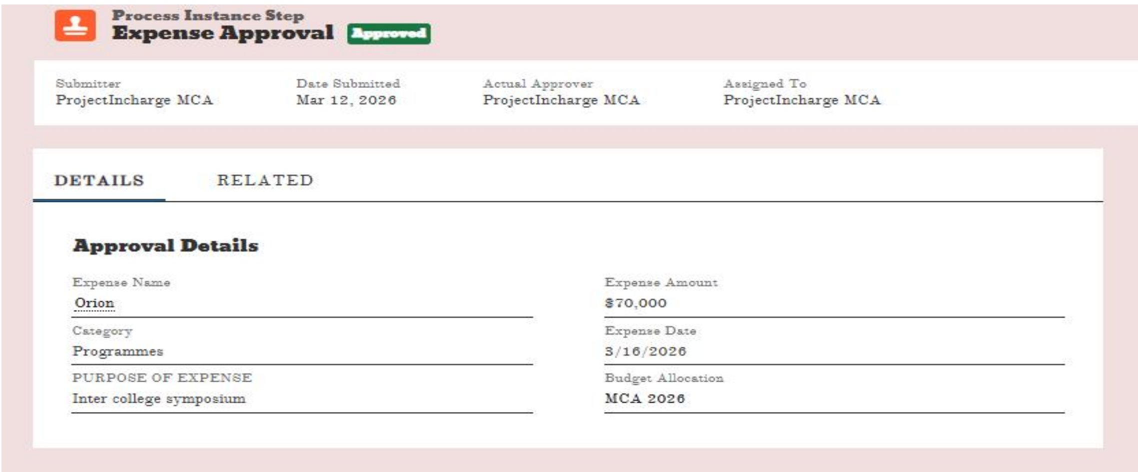


Figure B.10 Approved Expense Page

In Figure B.10 this screen shows the Approved Expense Details page of the Department Budget Tracking System. This page displays the details of an expense after it has been approved, including the expense name, category, purpose, expense amount, expense date and budget allocation. It confirms that the expense request has been successfully approved through the approval workflow.

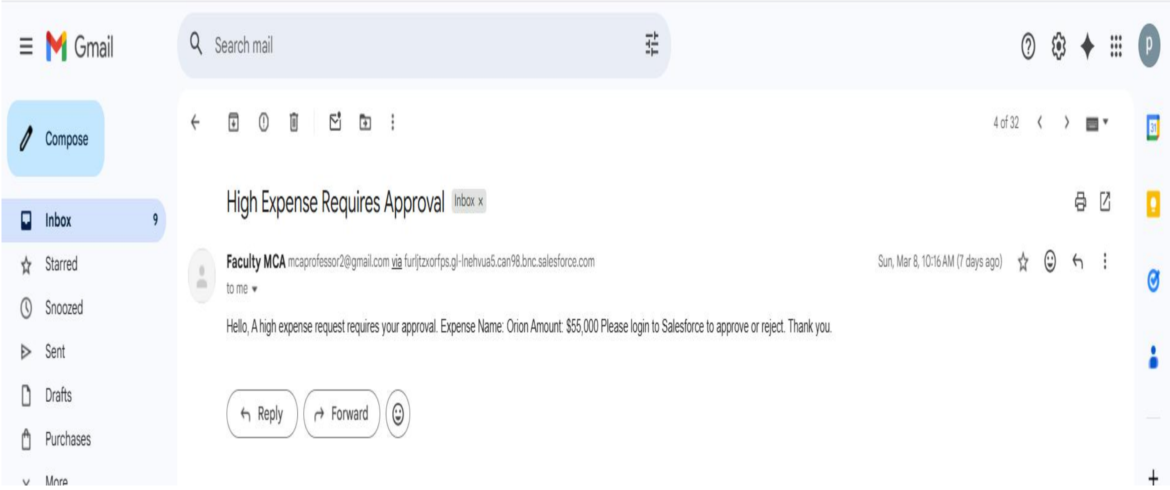


Figure B.11 Expense Approval Email Notification

In Figure B.11 this screen shows the email notification sent by the system when a high expense requires approval. The email informs the concerned authority that an expense request has been submitted and requires review. It includes important details such as the expense name and expense amount and instructs the approver to log in to Salesforce to approve or reject the request. This notification helps ensure timely approval and proper monitoring of departmental expenses.

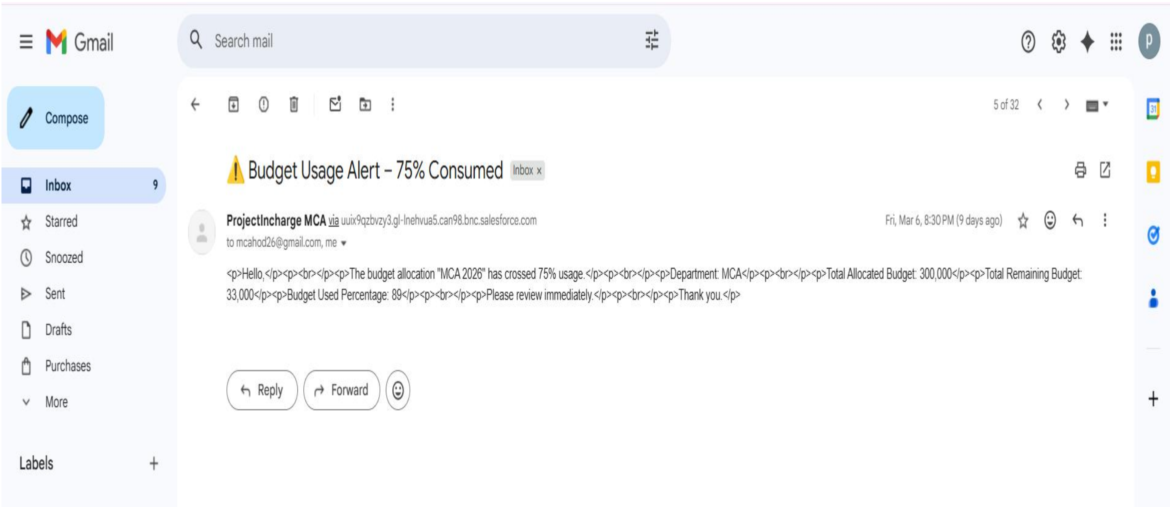


Figure B.12 Budget Usage Alert Email Notification

In Figure B.12 this screen shows the budget usage alert email generated by the system when the department budget crosses a defined usage limit. The notification informs the administrator that the budget allocation has exceeded 75% consumption. The email also displays important details such as department name, total allocated budget, remaining budget and budget usage percentage. This alert helps administrators monitor departmental expenses and take necessary actions to control further spending.

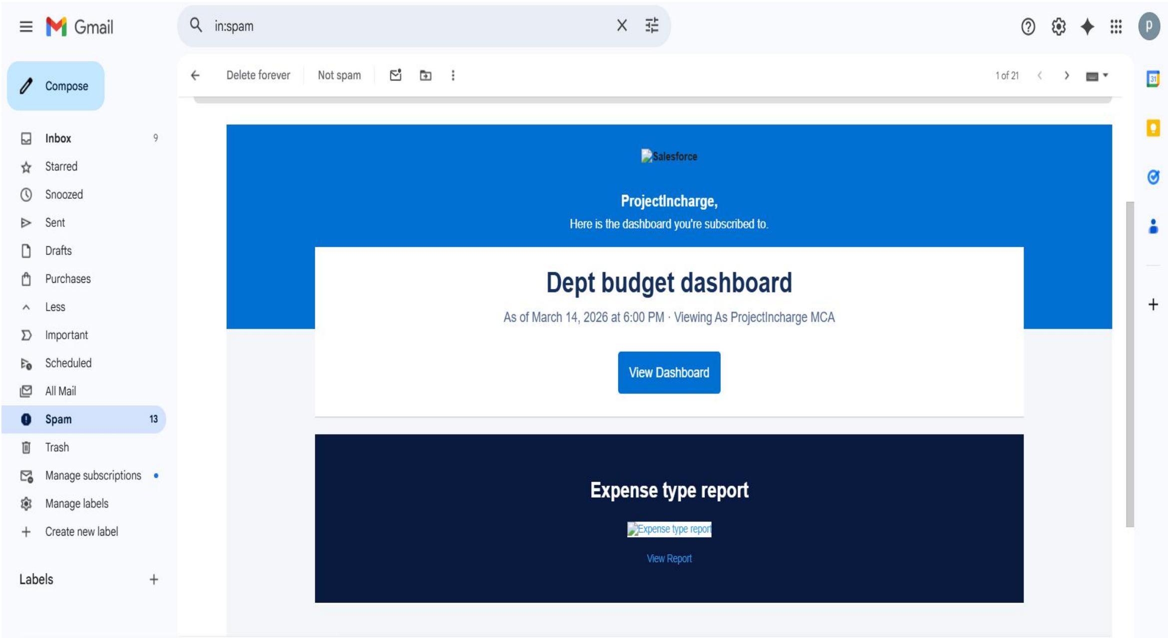


Figure B.13 Dashboard Subscription Email Notification

In Figure B.13 this screen shows the dashboard subscription email notification sent by Salesforce. The email notifies the user that the Department Budget Dashboard report is available for viewing.

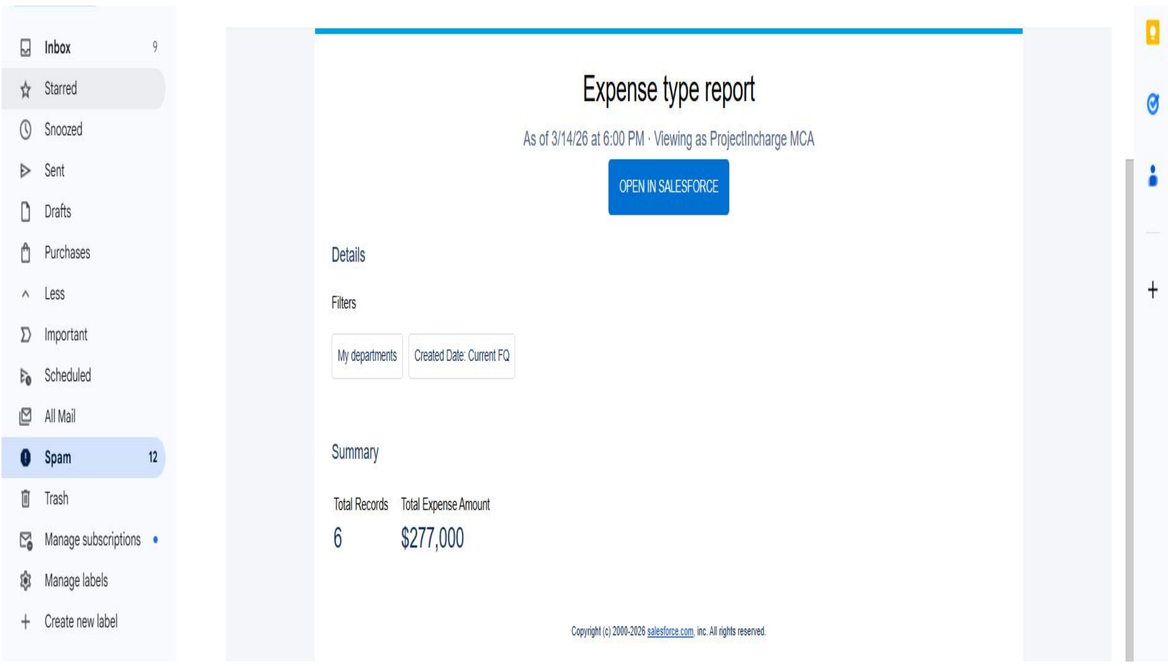


Figure B.14 Expense Type Report Email

In Figure B.14 this screen shows the Expense Type Report sent through email subscription from Salesforce. The email displays a summary of the expense report, including the total number of records and the total expense amount for the department.

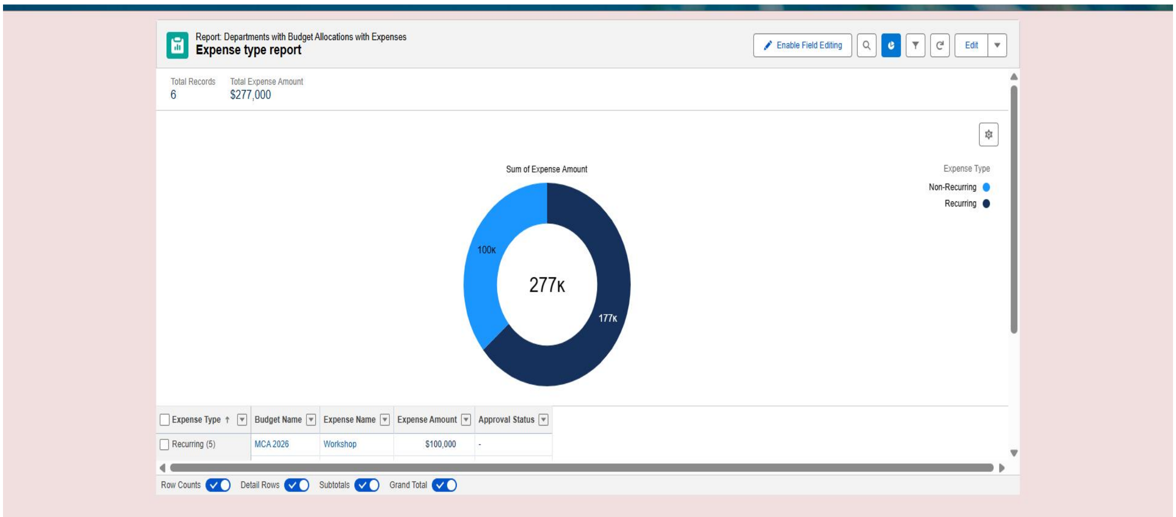


Figure B.15 Expense Type Report

In Figure B.15 this screen shows the Expense Type Report of the Department Budget Tracking System. The report provides a visual representation of the total expense amount categorized by expense type, such as recurring and non-recurring expenses.

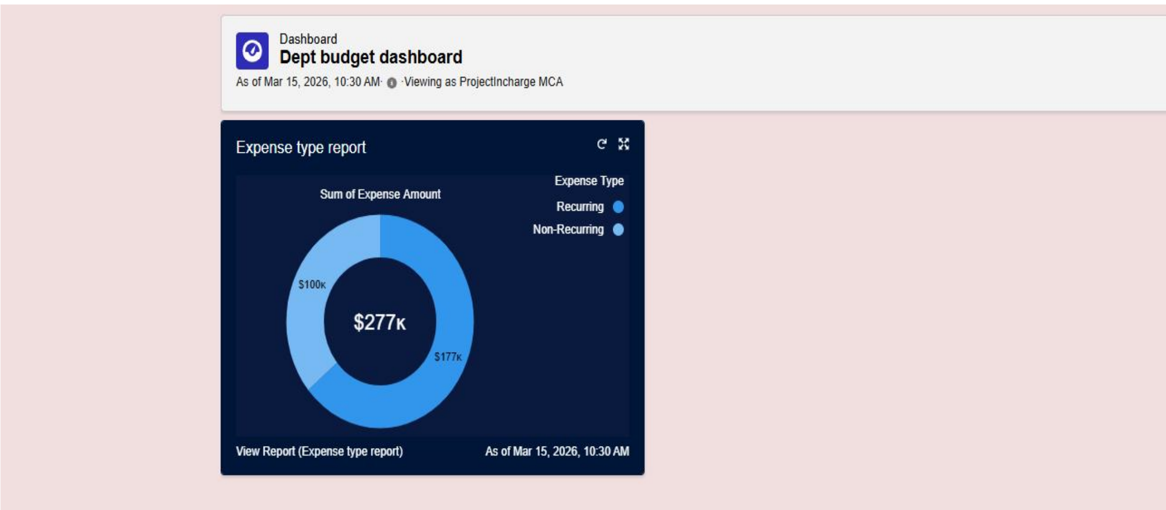


Figure B.16 Expense Type dashboard

In Figure B.16 this screen shows the Expense Type Dashboard of the Department Budget Tracking System.The dashboard provides a visual representation of the total expense amount categorized by expense type, such as recurring & non-recurring expenses.

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